

Central and Southern Asia

Research and Action Agenda

List of Authors

Sameen Nasar – BRAC James P Grant School of Public Health, , BRAC University, Dhaka, Bangladesh

Wafa Alam - BRAC James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

Md Tanzirul Alam - BRAC James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

Aishath Enash – U-Inspire Maldives and Ministry of Climate Change, Environment and Energy, Maldives

Poornima Prabhakaran - Centre for Health Analytics Research and Trends, Trivedi School of Biosciences, Ashoka University, India

Kunnuru Duishobekova - Kyrgyz-Turkish University Manas, Bishkek, Kyrgyzstan

Kezang Wangmo - BRAC University, Dhaka, Bangladesh

Inayat Singh Kabir - The George Institute for Global Health, India

Jawan Shir Rashik - Afghanistan Research and Evaluation Unit (AREU), Kabul, Afghanistan

Dr Farzana Misha - BRAC James P Grant School of Public Health, , BRAC University, Dhaka, Bangladesh

Table of contents

Executive summary	3
Introduction	4
CCM project overview	6
Methodology	7
Current state and emerging needs for climate change and mental health in Central and Southern Asia	12
Research agenda	18
Action agenda	27
Discussion: strengths, limitations and next steps for the research and action agenda	41
Conclusion	42
Hearing from the regional community of practice	44
Glossary of terms	45
Who produced this report	45
Appendix	49
References	62

Executive summary

Climate change is increasingly recognised as a threat to mental health, compounding risks for poor mental health outcomes and destabilising the conditions needed for good mental health. While research at the intersection of climate change and mental health has proliferated in recent years, the field remains disconnected, unequal and siloed, slowing urgent progress to address the mental health impacts of climate change and gain the mental health benefits of climate action.

Connecting Climate Minds (CCM) is a Wellcome-funded initiative to cultivate a collaborative, transdisciplinary climate change and mental health field with a clear and aligned vision. Over the last year, we have convened experts across disciplines, sectors and countries in virtual dialogues to develop regional and global research and action agendas. These agendas set out 1) research priorities to understand and address the needs of people experiencing the mental health burden of the climate crisis, and 2) priorities to enable this research and translate evidence into action in policy and practice. We hope this work will help to focus the efforts of funding, policy and practice communities working toward a future where no one is held back by climate-related mental health challenges.

This report presents the research and action agenda for climate change and mental health in Central and Southern Asia (CSA). The region – which, according to the Sustainable Development Goal regional groupings, includes Afghanistan, Bangladesh, Bhutan, India, Iran (Islamic Republic of), Kazakhstan, Kyrgyzstan, Maldives, Nepal, Pakistan, Sri Lanka, Tajikistan, Turkmenistan and Uzbekistan – faces increasing climate-related hazards such as floods, landslides, extreme heat and storms. These climate issues impact mental health through various mechanisms, including the destruction of livelihoods, infrastructure and social bonds. Furthermore, many countries in the region experience multiple burdens of climate disasters (e.g., prolonged drought and flooding), which leads to external and internal migration, compounding the effects on the mental health of individuals and their communities.

Cross-disciplinary research on climate change and mental health for this region is limited; this was evident through the insights and discussions that took place across the two CCM regional dialogues. Through the synthesis of data from the dialogues and surveys of the Regional Community of Practice (RCoP), we identified a total of 43 research priority themes. Some of the major topics include:

- **Unequal impacts of climate disasters** on the most vulnerable and neglected groups. These groups include children (under 5), students, urban slum dwellers/migrants, single female household heads, the elderly, urban poor, pregnant women, and adolescents.
- **Low levels of public mental health knowledge** create difficulties in addressing climate-induced mental health issues.
- **Effects of climate disasters** – such as migration (internal & external), adverse effects on physical health, destruction of households and communities, and the aggravation of household-level conflict and gender-based violence – lead to mental health challenges.
- **Developing a comprehensive bio-psychosocial model that is culturally sensitive** and can be applied to understand mental health experiences and challenges in the context of climate change.
- **Incorporating lived experiences** alongside traditional means of assessing mental health (psychometric measures).
- **Developing integrated interventions** that include researchers, governments and lived experience individuals, and where possible, enhance existing interventions to better reach marginalised groups.

To address these research topics, regional online dialogue participants emphasised the need for inclusive research, which strengthens community-based adaptation initiatives. This will require sustained collaboration efforts between researchers, governments, funders and those delivering interventions.

This regional research and action agenda serves a critical role in synthesising and disseminating many of the key issues in the climate-mental health space in CSA. Additionally, it sheds light on marginalised groups whose issues and opinions have not traditionally been considered within formal research. Furthermore, strengthening this RCoP has the potential to produce research outputs that prioritise vulnerable and marginalised groups within the region, enabling the representation of lived experiences at all levels; this will eventually strengthen existing and emerging interventions for climate-related mental health challenges.

Introduction

Context

Climate change and mental health are two of our greatest global challenges and awareness of the intersection between mental health and the climate crisis has grown rapidly in recent years.¹ Climate change exacerbates mental health challenges by increasing exposure to extreme heat and the traumas of extreme weather events,² destabilising the conditions needed for good mental health and wellbeing (for example, water and food insecurity, forced migration, polluted air, loss of treasured environments),² disrupting access to healthcare³ and increasing psychological distress through awareness of climate threats and insufficient climate action.⁴ People living with mental health challenges are also particularly vulnerable to the stressors of the climate crisis, such as increased risk of physical heat stress and death during heatwaves.^{5, 6, 7}

In response to the mounting mental health toll of the climate crisis, research in the climate and mental health field has grown rapidly. Nevertheless, key evidence gaps exist in many regions, including the mental health burden attributable to climate change, the pathways and mechanisms underlying these impacts, the co-benefits of climate action for mental health and the best interventions or solutions to support mental health in a changing climate. Climate change and mental health research remain frustratingly disconnected across disciplines, sectors and geographies, and are unevenly focused on certain topics and global regions.⁸ Moreover, siloed decision-making slows the translation of evidence to aligned action across climate and mental health policy and practice.^{9, 10} A more inclusive, connected agenda is urgently needed to generate the evidence to truly understand, monitor and respond to the interconnections between climate change and mental health.

Connecting Climate Minds

Connecting Climate Minds (CCM) is a Wellcome-funded project launched in 2023 to develop an inclusive agenda for research and action in climate change and mental health. The project has two key, intertwined aims. The first is to develop an aligned and inclusive agenda for research and action that is grounded in the needs of those with lived experience of mental health challenges in the context of climate change, to guide the field over the coming years. The second is to kickstart the development of connected communities of practice for climate change and mental health in seven global regions (designated by the Sustainable Development Goals), equipped to enact this agenda. We aim to combine the strengths of a global perspective and regional focus and bring together diverse disciplinary perspectives into a shared vision that can ensure research is effective at addressing priority evidence gaps and informing changes in policy and practice at the intersection of climate change and mental health.

Through bringing together experts across diverse disciplines, sectors and countries, the CCM team has facilitated the development of a lived experience-informed research and action agenda for the climate change and mental health field in Central and Southern Asia (CSA).

Objectives of the research and action agenda

The research and action agenda is designed to focus future efforts to help those who are experiencing or will experience, the compounding mental health challenges of climate change. It aims to support those who are already responding to these challenges – through communities, research, policy and practice – by building a more connected and collaborative climate change and mental health field. It also aims to empower experts across disciplines and sectors to join and make progress in this area by identifying clear priorities and fostering a more inclusive and transdisciplinary field.

The agenda addresses these aims through three core objectives, which are to:

1. Identify priorities for research that can inform action to meet the needs of people experiencing and responding to the mental health impacts of climate change in CSA.
2. Identify what is needed to appropriately conduct research and translate evidence to action in policy and practice in CSA.
3. Build understanding among researchers, funders and policy experts across disciplines and sectors of their role in furthering climate change and mental health research and equip them with these clear and actionable priorities.

The regional agenda will be integrated with six other regional agendas to inform a **global research and action agenda** for climate change and mental health. This will ensure that global research efforts and investment in climate change and mental health are grounded in regional-level priorities. Importantly, the global agenda will also integrate insights from agendas developed with and for some of the most affected groups globally, namely Indigenous Peoples, youth, small farmers and fisher peoples.

Use of the terms climate change and mental health

Climate change, mental health and their intersection are complex and wide-ranging fields. For the purpose of this agenda, we define the scope of these terms as follows:

By **mental health challenges**, we mean thoughts, feelings and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes, but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts and substance misuse.

By **experiences of the effects of climate change**, we mean: 1) experiencing direct impact of climate hazards, such as more frequent and intense heatwaves, wildfires/bushfires, drought, floods or storms (e.g., typhoons, hurricanes, cyclones), and 2) experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices because of climate change.

Mental health challenges in the context of climate change include:

- How climate change may lead to worsening pre-existing mental health challenges.
- How climate change may contribute to the prevalence or impact of existing mental health challenges.
- How climate change may impact treatment access or effectiveness for those with mental health challenges.
- How climate change may lead to new mental health challenges.

The convening work of CCM presents a key opportunity to build our understanding of diverse perspectives, framings, and terminologies in CSA, which we have sought to reflect within the research and action agenda.

CCM project overview

Regional Community Team

In CSA, CCM is led by a Regional Community Team (RCT), responsible for convening diverse expertise across the region and building regional capacity to create and enact the research and action agenda. The structure of the RCT is outlined below.

Regional Community Convenor (RCC)

Purpose: Responsible for developing and delivering project activities in the region, including convening and supporting a regional community of diverse expertise.

Members: BRAC James P Grant School of Public Health, BRAC University, Bangladesh:

- Dr Farzana Misha
- Wafa Alam
- Sameen Nasar
- Md. Tanzirul Alam

Co-convenors

Purpose: Bringing an additional breadth of expertise across disciplines and countries (i.e., organisations spanning climate expertise, stress neuroscience and mental health expertise, in different sectors), providing technical advice and review and supporting project delivery.

Members:

- Roufa Khanam, Assistant Director, Centre for Climate Change and Environmental Research (C3ER), BRAC University, Bangladesh
- Zul Merali, Founding Director, Brain and Mind Institute, Aga Khan University, Pakistan

Lived Experience Advisory Group (LEAG)

Purpose: Advisory board of experts with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. Drawing on their unique expertise and wisdom, LEAGs provide vital community-centred perspectives and guidance that inform the overarching approach and outputs of the project.

Members:

- Monira Rahman, Founder, Innovation for Wellbeing Foundation, Bangladesh
- Poornima Prabhakaran, Director, Centre for Health Analytics Research and Trends, Trivedi School of Biosciences, Ashoka University, India
- Matrika Devkota, Founder and Executive Director, KOSHISH, Nepal

Youth ambassador (YAs)

Purpose: Youth advisors (aged 18-29) with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. YAs bring unique youth-centred perspectives to the development and implementation of project activities.

Members:

- K.M. Miraj Sujoy, Student, BRAC University, Bangladesh
- Kezang Wangmo, Student, BRAC University, Bhutan

- Kunnuru Duishobekova, Student, Kyrgyz-Turkish University Manas, Kyrgyzstan
- Aishath Enash, Environmental Manager, Board Member, U-Inspire Maldives and Program Officer, Ministry of Climate Change, Environment and Energy, Maldives

Methodology

We produced this research and action agenda through a robust and inclusive methodology to capture, combine and refine a rich diversity of perspectives while fostering connection across a growing community of practice. The CCM core team developed this methodology in consultation with RCTs, a Global Advisory Board and Wellcome. Methods and materials were adapted regionally to ensure a balance of global standardisation with regional appropriateness and flexibility. Continuous sharing between regions of processes, learnings and challenges facilitated the iterative development of the methodology. The process for developing the regional research and action agendas is shown below.

Figure 1: Research and action agenda development methodology



Pre-dialogue scoping

Global scoping and framing

The global core CCM team performed a global-level scoping review of current reviews, key papers and policy reports relating to the climate change and mental health field. They undertook a mapping of the research categories covered by and used to structure previous reviews of the field, followed by a second mapping of the recommendations for action on climate change and mental health proposed by reports written to inform policy and practice. These results have been used to frame the dialogues and research agenda to align with the current field while responding to key gaps.

Regional scoping

A regional pre-dialogue scoping was conducted to understand how the intersection of climate change and mental health was conceptualised across the different countries, the cultural context and the lived experiences of individuals across the region. This information was summarised in “Mental health challenges emerging in Central and Southern Asia in the context of climate change” and shared in the pre-reading materials with dialogue participants for contextual understanding before the first dialogue.

The CSA team conducted a rapid literature review on climate change and mental health in the region. The first search was conducted using the following keywords “Climate change and mental health” AND “Central

and South Asia". A second search was conducted using more specific keywords: "Climate Change in South Asia"; "Climate Change in Central Asia"; "Mental Health in South Asia"; "Mental Health in Central Asia". After review, the two searches were narrowed down to 20 peer-reviewed articles and grey literature. In addition, dialogue participants provided articles and abstracts on work and research being conducted in their respective countries. These were used to prepare pre-reading materials for the dialogues; some of the literature was also used in the context setting for this research and action agenda.

During the initial meetings and discussions with potential dialogue participants and individuals, several stakeholders expressed interest in being part of the CSA regional community of CCM, including Afghanistan Research and Evaluation Unit (AREU), Vihara (India), Huesofthemind Mental Wellbeing Foundation (India), Dr. Lahiru Kodithuwakku (UNFPA Sri Lanka) and Dr. Wais Aria (Tabish Social Health Education Organization, Afghanistan), and shared their publications and working papers. These were used to develop the pre-reading materials as well as the regional dialogue presentation.

Dialogue methodology

Participant Selection

For the first two months of the project, the CSA RCT conducted brief (20 - 30 minutes) introductory meetings with potential stakeholders from our existing network of individuals and organisations. These include partner organisations of BRAC James P Grant School of Public Health (JPGSPH), such as the George Institute for Global Health, and JPGSPH's Master of Public Health programme's alumni networks working in mental health and/or climate change. Connections were built to learn about their work at this intersection of climate change and mental health, as well as their prospective participation in the CSA Regional Community of Practice (RCoP) for CCM. Additional participants were recruited from individuals who registered interest in becoming part of the CSA RCoP via the Qualtrics form on the global website.

A web search was conducted by the CSA team to identify relevant research organisations' international, local, and grass-roots NGOs; government bodies; and civil society organisations. Introductory emails were sent to these organisations and institutions for an initial meeting to identify potential dialogue participants. Recommendations for dialogue participants were also shared by our co-conveners, LEAG and other confirmed participants, whom we contacted over email.

Dialogue preparation

Dialogue agendas were designed by the global team, amended to be locally appropriate and facilitated by members of the RCoP, who are themselves research, policy and lived experience experts. Facilitators and note-takers, who were recruited from BRAC JPGSPH, were provided a brief one-hour orientation on dialogue facilitation, where their roles were explained and relevant documents were shared with them. All facilitators and note-takers were also required to take an online training course for psychosocial first aid to support participants who might experience distress during the session.

Dialogue sessions

There were two dialogues (3 hours each) with experts across disciplines, sectors, and countries in CSA (see Appendix 1 for dialogue agendas). Both dialogues were conducted in English. Although multiple languages are spoken across the CSA region, English is widely used and understood. We conducted separate meetings with participants who did not understand English to gather their feedback on dialogue topics. Many facilitators actively contributed to the discussions and therefore played a role in both data collection and production. The breakout sessions were audio-recorded using the recording feature on Zoom, and the recordings were shared with ICL.

- **Dialogue 1:** In the first dialogue, participants identified regional needs and research priorities. There were eight breakout rooms in Dialogue 1, each with one facilitator and one note-taker. The breakout rooms were designed to have representation from different sectors and countries, except one specifically for Bengali speakers. All facilitators and note-takers used Google Docs for notetaking.
- **Dialogue 2:** The second dialogue gathered feedback on draft research priorities, identified strategies to undertake research in the region and translate evidence to action in policy and practice, and explored the diversity of regional perspectives and understandings of key concepts. Dialogue participants were asked to refine the research priority themes/questions identified in Dialogue 1. This was done using Google Jam board. For each of the research questions, the following prompts were used to lead the discussion:
 - *Is this a valuable theme? Would you add or change anything?*
 - *Share any ideas on datasets, metrics or methods that could be applied to help address this.*

A spectrum mapping exercise was also conducted during the dialogue, where participants shared their perception and understanding of the following concepts: climate vulnerability, climate action and mental health.

Survey methodology

1. Pre-dialogue survey:

The pre-dialogue survey was sent to all invited dialogue participants to collect information on their socio-demographic characteristics, personal and professional experiences with relation to climate change and/or mental health issues, and perceptions on the intersection of climate change and mental health. The survey also had questions on accessibility requirements to ensure that any special needs from the participants (such as language preference, subtitles or translation support) could be met.

2. Post-dialogue survey:

A post-dialogue survey was developed a week after the second regional dialogue. This was sent to participants who attended either or both the regional dialogues. The post-dialogue survey was also sent to a wider regional community including those who expressed interest to receive project updates.

Analysis methodology

Analysis of Dialogue 1 was conducted by core members of the CSA RCT using the Framework Method¹¹ – a matrix-based approach that allows qualitative researchers to undertake deep interrogation of transcripts and written notes. The analyst team populated a matrix based on the global coding framework by creating nuanced summaries with key quotes drawing from the notes and transcripts of the breakout discussions.

A draft list of research priority themes/research questions was prepared using data from Dialogue 1. These were refined based on scoping guidance from Wellcome and consultation with the RCoP and triangulation across breakout notes, transcripts, pre-dialogue scoping and expert opinion. The draft research themes were then shared with Dialogue 2 participants and a wider range of experts through the post-dialogue survey. Based on the feedback received from participants in the dialogues, post-dialogue surveys, and Wellcome, the RCT finalised the list of research priorities for the region (see Table 5).

The action agenda was prepared based on discussions in the breakout sessions of Dialogue 2, using a similar framework method for analysis. Coding frameworks for the research agenda and action agenda can be found in Appendix 2.

Participants

Dialogue participants were a diverse group across geographical spread, gender, sector and discipline. All participants were invited to both dialogues; however, in some cases, participants were unable to attend both dialogues.

In total, 65 participants attended Dialogue 1 and 56 participants attended Dialogue 2. The tables below provide a breakdown of participant characteristics.

Table 1: Geographical spread

Country	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Afghanistan	4	67%	3	9%
Bangladesh	25	42%	21	60%
Belgium	0	0%	1	3%
Bhutan	1	2%	0	0%
India	8	13%	5	14%
Iran (Islamic Republic of)	1	2%	0	0%
Kazakhstan	1	2%	0	0%
Kenya	0	0%	1	3%
Kyrgyzstan	1	2%	0	0%
Maldives	3	5%	0	0%
Nepal	4	7%	1	3%
Pakistan	6	10%	1	3%
Sri Lanka	4	7%	2	6%
United Kingdom of Great Britain and Northern Ireland	1	1.7%	0	0%
Uzbekistan	1	1.7%	0	0%

Table 2: Expertise

Dialogue 1		Dialogue 2	

Expertise	Number	Percentage	Number	Percentage
Climate Change	38	38%	25	44%
Mental Health	25	25%	12	21%
Health	27	27%	16	28%
Other	7	7%	2	4%
I do not know / Prefer not to say	3	3%	2	4%

Table 3: Discipline

	Dialogue 1		Dialogue 2	
Discipline	Number	Percentage	Number	Percentage
Activism	20	9%	11	10%
Community	43	20%	21	18%
Education	20	9%	9	8%
Expert through my own lived experience	12	5%	5	4%
Funding	4	2%	0	0%
Healthcare	14	6%	7	6%
Non-governmental Organisations	43	20%	21	18%
Policy	20	9%	14	12%
Research	38	17%	26	23%
Other	6	3%	1	1%

Table 4: Gender

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage

Men	19	33%	8	35%
Women	37	64%	13	57%
Non-Binary	2	3%	1	4%
I do not know / Prefer not to say	0	0%	1	4%

Survey participants*

Pre-dialogue survey: 85

Post-dialogue survey: 30

**Please note numbers are approximate and do not account for duplicate or incomplete responses*

Ethics, data collection and storage

This study has been reviewed and given an ethically favourable opinion by the Imperial College Research Ethics Committee (Study title: 'Global Dialogues to set an actionable research agenda and build a community of practice in climate change and mental health', study ID number: 6522690).

Given the prior ethical approval from ICREC and the scope of the project, no additional approval was required from the BRAC JPGSPH.

Emphasising transparency, information sheets assured participants that collected data is exclusively for research, with the published report safeguarding confidentiality. Contact details were provided for queries about the consent process, demonstrating a commitment to ethical considerations throughout the study's lifecycle.

Details on data collection and storage can be found in Appendix 5.

Current state and emerging needs for climate change and mental health in Central and Southern Asia

As the climate crisis escalates, more people globally are experiencing related mental health consequences. However, the current evidence base doesn't fully capture these experiences. To develop research themes that ultimately meet the needs of those experiencing and responding to the interconnections between climate change and mental health, it is vital to know: 1) what do people from different backgrounds, contexts and sectors – particularly those with lived experiences of mental health challenges in the context of the climate crisis – report as their experiences needs, and resiliencies? and 2) what evidence do people making decisions and taking actions on the ground need to adequately respond?

This section sets out the context of the research agenda, presenting a synthesis of what we heard through the dialogues, surveys, expert consultations and literature review as key emerging needs for mental health in the context of climate change. We also present regional perceptions and insights from

the literature on the current state of evidence to explore which key emerging needs are not yet being captured through research.

Current research on climate change and mental health in Central and Southern Asia

Quality studies using appropriate methods and metrics are required to better understand the full picture of climate-related mental health impacts in many regions globally.⁸ This remains the case for CSA. Although climate-related hazards such as floods, landslides, extreme heat and storms strongly impact the CSA population, little is understood about the climate-induced effects on mental health. This is partly due to the low levels of mental health knowledge among the public within the region.¹² Despite this lack of mental health literacy, many South Asian countries have some of the highest rates of common mental health disorders globally.¹³ Concerning countries in Central Asia, mental health needs are under-researched;¹⁴ while some efforts have been made to address mental health issues, stigmatisation prevents further research efforts. Together with an insufficient allocation of resources and low prioritisation by governments, stigmas and psychiatric myths persist within Central Asia.¹⁵

Vulnerabilities associated with climate events in the region include socioeconomic status (with higher risk for low socioeconomic status), age (higher risk for very young and very old people) and those with pre-existing health conditions. Climate-related mental health impacts among these groups can arise from food insecurity, lack of water access and migration.¹⁶ While there is scant literature making the links between climate events and mental health impacts in the region, such climate impacts and individual vulnerabilities have been negatively associated with mental health and wellbeing in other regions.¹⁷ Other relevant documented issues within the region include forced migration, increases in women's responsibilities, extreme heat and its impact on psychological burden. Gender-based violence is also prevalent; it has been established that hot temperatures can increase aggressive motives and behaviours.¹⁸ Furthermore, the severity of climate events has led to increasing farmer suicides; suicides have been documented amongst farmers in India and Bangladesh.^{19, 20}

Concentrated efforts are required to address the gaps in research and action on climate-induced mental health challenges concerning differing cultures, as well as neglected vulnerable groups.

Framing of key concepts

The overall scope and focus of CCM have been guided by the framing of climate change and mental health outlined in the introduction. 'Climate change', 'mental health' and their intersections and related terms, along with other relevant key concepts, are also understood and defined in diverse ways in CSA. Identifying, acknowledging, and honouring the ways these terms are understood and used in different settings is critical to help foster connections, awareness and recognition across disciplines, cultures and communities. This section highlights regionally relevant understandings, as generated through a mapping activity in Dialogue 2, designed to reveal the diversity of perspectives around key concepts and other discussions throughout the project.

Mental health

In CSA, the term 'mental health' is often defined holistically, considering psychological, emotional and social aspects. Cultural influences connect mental health with spirituality or communal ties. One of the key aspects of mental health is having control over one's feelings and emotions. Like the global understanding of mental health as a comprehensive state of wellbeing, it also recognises specific mental health conditions such as anxiety, depression and PTSD as major mental health challenges that hinder a person's ability to function in one or more spheres of life. In the context of CSA, there exists stigma surrounding mental health that often hinders open communication. Socioeconomic disparities among

communities and limited mental health services lead to complexities in addressing the growing crisis of mental health challenges in the region.

Climate vulnerability

The term 'climate vulnerability' primarily refers to the challenges faced by diverse communities in the region due to climate change effects such as sea-level rise, cyclones, flooding and the resulting disruption to agriculture and other economic and social activities. Climate vulnerability can be a broad term encompassing vulnerability to climate risks and hazards, geographic vulnerability (coastal regions are an example), economic vulnerability and disease vulnerability. Beyond the evident environmental and physical impacts of climate change in CSA, vulnerability extends to various socioeconomic factors. Communities already living in poverty, with limited access to education and social marginalisation, bear a disproportionate burden. The loss of agricultural land, livestock and income sources due to climate-induced events further magnify the precarious nature of livelihoods. Climate vulnerability is also interlinked with education, hindering children's access to drinking water, food, sanitation and hygiene, and disrupting their educational pursuits. In the context of mental health, the voices of those already vulnerable are often unheard, aggravating their susceptibility to climate-induced hardships and contributing to a sense of helplessness and distress. Overall, the framing of climate vulnerability in CSA reflects the interconnectedness of climate risk, economic disruptions, health issues, societal challenges, and the mental health dynamics of already vulnerable populations.

Climate action

The definition of 'climate action' extends beyond efforts to directly combat climate change; it includes actions to mitigate the impact of climate change on mental health. Participants from countries across the region agree that every climate-vulnerable population should have access to mental health support, acknowledging the mental health challenges that may not always be visible but are deeply impactful. Religious leaders are recognised as influential actors in climate action, highlighting the need for diverse stakeholders' involvement, including but not limited to women, children, farmers and other community members. In terms of climate action, awareness and education of climate change impacts, including on mental health, is important and is required for informed decision making and innovative solutions thinking. Overall, the understanding of climate action in CSA reflects a holistic approach that integrates mental health considerations, community resilience, policy changes and widespread awareness initiatives.

Regional needs for mental health in a changing climate

While the climate crisis affects every part of the world, current and future hazards are unevenly distributed and variable. For instance, coastal communities may face increasing rises in sea level, coastal erosion and storm surges, while Inuit communities face melting sea ice. Different parts of the same country or region may face different levels of threat to droughts, wildfires, floods or extreme heat. Understanding the predicted climate hazards and their timescales for different communities and regions is vital to appropriately target support for mental health.

This section outlines the key climate exposures facing CSA projected to approximately 2030, as modelled by climate experts at the Red Cross Red Crescent Climate Centre. We also present findings from the dialogues, surveys and consultations with lived experience experts, co-conveners, youth ambassadors and others members of the RCoP to explore perceptions of: 1) mental health risks associated with these climate exposures, 2) whose mental health may be most at risk, 3) the pathways through which climate exposures might produce or exacerbate existing mental health challenges, 4) climate adaptation and mitigation actions that may benefit mental health, and 5) mental health actions/solutions that can help respond to these mental health impacts.

Climate hazardsⁱ

As a region, CSA is particularly vulnerable to climate change. This region has frequently experienced droughts, wildfires, floods, landslides, storms and extreme temperatures over the recent years, as international disaster database [EM-DAT](#) sources show. The region is facing an increase in the frequency and severity of a range of climate hazards, including the following, modelled to approximately 2030 as compared to the historical baseline (generally 1986-2005):^{21, 22}

- **Extreme heat**, particularly across the western parts of CSA, with an increase of more than 20 hot days per year in Iran, Afghanistan, Turkmenistan and southern India (high confidence);ⁱⁱ
- **Heavy precipitation events and flooding** east of the Caspian Sea in Kazakhstan and India (high confidence); and
- **Sea level rise** across the region, resulting in more coastal flooding (high confidence).

In addition to the listed climate challenges, countries in the Iran-Afghanistan border also experience dust storms. The adverse effects of these climate-induced impacts range from impacts on agriculture, disruption of basic services and forced migration.²³

All of these climate hazards have compounding effects. Floods and landslides destroy homes and community infrastructure, where salinity intrusion due to sea-level rise destroys cultivated land, leading to crop failures across farmlands in the region. Increased temperatures lead to extreme heat and drought, where some urban areas experience heat island effects, and other areas experience the drying up of water bodies which affect surrounding forestry and increase dust emissions.

In CSA – in countries like Kazakhstan, Kyrgyzstan and Tajikistan – the level of air pollution in large cities far exceeds WHO standards.²⁴ The air in these cities contains suspended particles of PM2.5, which can penetrate different organs and the brain causing multiple disorders. In Almaty, Kazakhstan, according to the National Hydrometeorological Service of the Republic of Kazakhstan, average annual pollution levels are 2.7-8.1 times higher than WHO limits.²⁵ Bishkek in Kyrgyzstan regularly reaches record-breaking pollution levels during heat seasons. The main contributors to air pollution are transport, cutting down trees, disruption of wind patterns and heating from non-renewable energy sources (such as coal and easily combustible materials) in residential areas on the outskirts of the city, where there is no access to central heating.

What mental health outcomes appear to be impacted?

The following mental health challenges were raised as associated with climate change:

- Bipolar disorder
- Depression and apathy (particularly associated with seasons of heavy air pollution and smog)
- Neurological disorders (e.g., Alzheimer's disease, Parkinson's disease and multiple sclerosis)
- Panic attacks
- Post-traumatic stress disorder (PTSD)
- Schizophrenia
- Stress and anxiety
- Suicidal ideation

ⁱ Information based on Red Cross Red Crescent Climate Centre mapping exercise and projections (methods detailed in Appendix 6). Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the Coupled Model Intercomparison Project - Phase VI (CMIP6) multi-model GCMs ensemble provided in the IPCC, 2021.

ⁱⁱ Within the IPCC and other major sources of climate projections, confidence levels are given on a scale of low, medium, high. The ranking refers to the robustness of the evidence available and the agreement between climate models.

In the context of CSA, mental health effects are mostly described and understood by laypeople in terms of changes in emotional states. As such, the following were also raised:

- Emotional variability including sadness
- Heat-related stress and discomfort

Who is particularly affected?

- Agricultural workers (farmers, fisher peoples) and their families
- Children
- Coastal communities
- Lower socio-economic groups
- Migrants and those in temporary, remote, or insecure housing
- Outdoor workers
- Students
- The elderly
- Urban poor or people living in informal settlements
- Women

What are the pathways and mechanisms linking these climate hazards to mental health outcomes?

- **Economic insecurity:** Climate disasters can have negative effects on household economic situations. Destruction of infrastructure and transport lead to reduced job seeking opportunities and therefore reduced income and resulting mental stress.
- **High dependence on agriculture and fisheries for livelihood:** Households reliant on agriculture and fisheries, particularly in coastal regions of the Maldives, India and Bangladesh, are severely affected by climate-induced disasters. These communities witness the destruction of their livelihoods, often leading to migration in search of alternative income sources. There is also a rise in single female-led households, with women taking on both household and income-earning responsibilities during periods of climatic distress. Women, primarily responsible for marketing and sales in the fishing industry, face reduced earnings due to climate impacts, exacerbating stress levels and care-giving burdens, which has ramifications on mental wellbeing, care-giving roles and responsibilities, thereby setting off a vicious cycle.
- **Loss and damage to infrastructure:** Mental health issues, such as PTSD, can stem from the debilitation of community infrastructure (e.g., roads, water supply), as affected populations witness their livelihoods and families being destroyed. For example, following the sudden floods in Pakistan in 2022, two to three provinces were destroyed, leading to displacement of populations. For these communities, the trauma of witnessing flooding of their households and their lost livelihoods led to prolonged anxiety and stress.
- **Displacement and relocation:** Climate-induced migration to recover livelihood opportunities lost due to the effects of climate hazards is prevalent in this region. Among those that migrate, there are issues around livelihood and income generation, as migration can affect earning opportunities. This can lead to extended feelings of anxiety amongst, particularly for household heads, who often bear the responsibility as principal earners for the family.
- **Violence and conflict due to climate-induced migration:** Climate-induced migration can result in families being separated from their local support systems, placing a greater burden on adults in the household. This can lead to conflicts within the household, which are often a result of limited resources and reduced livelihood opportunities. According to participants, this often creates an

environment where aggression and conflict are predominant; resulting gender-based violence may result in mental health issues for women. Household economic pressures further aggravate domestic violence within families, leading to mental distress and suicidal ideation in parents and children.

- **Air quality reduction and neurological effects:** Participants highlighted myriad ways that air pollution can impact mental health, including: air pollution correlation with conditions such as depression, anxiety, bipolar disorder and schizophrenia; links between prenatal exposure to air pollution and neurological diseases like autism spectrum disorders; increased presence of psychological disorders, depression and apathy during periods of heavy smog; and increases in the density and diversity of bacteria and plant pollens, which can diminish quality of life and contribute to mental distress.
- **Immature or insufficient government action and aid programmes:** Dialogue participants noted that most government institutions and climate-related schemes in the region are at an inaugural stage of implementation. As such, these programmes that don't adequately prevent or address climate-related mental health challenges.
- **Limited services and capacities for mental health:** In many countries across the region, mental health services remain sub-optimal for addressing climate-induced mental health issues. They do not sufficiently build resilience in communities or account for the multitude of issues that are associated with climate-induced disasters, including mental health challenges, that unaddressed can increase the risk of poor mental health outcomes. Climate related schemes/interventions in the region are often ad-hoc and address disasters as they arise (floods, extended droughts and post-disaster reconstruction).
- **Heat-related complications:** Heat-related stress, particularly in crowded urban areas, can cause restlessness and aggressive behaviour. In many areas in the region, there is limited cooling and shade, worsening the mental health impacts of heat – particularly for vulnerable groups.

What climate adaptation and mitigation actions have benefits for mental health?

Disaster management responses

In the Maldives, an Island Based Disaster Management system is being developed. This aims to assess capacity of each island to respond to climate disasters and identify areas and communities vulnerable to climate disasters. Through this initiative, the government will provide aid and assistance to build human and technical disaster response capacity of each island. However, the majority of the islands do not have the equipment and capacity to deal with such events. For example, flood-vulnerable communities are identified and provided with instruments for dewatering, and in addition, they are provided with training on first aid and psychological needs. However, in islands that lack of technology or capacity, National Disaster Management Authority, Maldives National Defence Force and Red Crescent Society are activated to provide help for disaster services as well as psychosocial health issues.

Heat action plans and infrastructure initiatives

In some states in India, the government has developed heat action plans, which mandate the use of heat-resistant colouring (white colours with specific pigmentation to resist sunlight on urban roofs). These initiatives are designed to create cooler internal building environments, as they counteract the effects of increased sunlight exposure and heat. This can help workers function better in extreme temperatures, indirectly addressing factors affecting mental health. However, the implementation of these plans faces public policy barriers.

Some communities have built small dams with their own finances. The government is looking to build canals in Northern Afghanistan to mitigate effects of drought. Other government initiatives include the subsidisation (approximately 5 INR for a 2km trip) of electric air-conditioned buses in South Mumbai. The subsidies may reduce some of the financial burden and thereby have a positive effect on livelihoods and mental wellbeing.

What mental health actions can help respond to these mental health impacts?

Multiple delivery methods for climate-mental health services

Mental health actions and interventions in this region range from disaster and post-disaster interventions to those delivered by health workers, including midwives and community health workers (CHWs). During the pandemic and lockdown there was a psychological helpline in Pakistan called 'We Care', but such initiatives have not been continued nor are they standard across the region. One of the dialogue respondents from Pakistan reported that psychosocial support is completely missing from any sort of rescue activities taking place.

Grassroots interventions and youth volunteers

'Uthan Boithak,' which means yard meetings, involves community youth volunteers who arrange gatherings if someone in the community wants to talk about mental health issues. In Sri Lanka, health facilities have public health midwives who are trained in psychosocial support for crisis times. In the Maldives, the Red Crescent Society has a major role in providing psychological assistance for people post disaster.

Addressing mental health through health system development

One of our participants mentioned that their country's health minister is planning for a mental health hospital. However, the general public have very little knowledge on mental health and associated stigmas would prevent them from seeking care.

"I talked with the health minister last time. So, she told me that there is a mental health hospital coming up in the future, but I asked her, though we have a mental health hospital in our country, who are we focusing on? People don't know about mental health hospitals coming up and they feel terrible to go to this kind of hospital." (Participant, Dialogue 1)

In government hospitals in the Maldives, free psychiatric or psychologist consultations are provided; though this leads to long queues, as there aren't enough clinicians to meet demand. Participants have noted that using community health workers, community midwives or other health workers may be disadvantageous. Primary commitments of health workers can lead to inefficient application of mental health services. Others have suggested more systemic interventions that would address mental health distress from a prevention standpoint. This could include promoting and strengthening existing mental health services in a country's respective health system.

Research agenda

Priority research themes

Background to research categories and priority research themes

This research agenda presents an aligned vision to guide the climate and mental health field in CSA. Research priorities have been generated through consultation with experts across disciplines, sectors and geographies in the region and iterated with experts regionally and globally. The priority research themes represent areas where targeted research investment could create a full picture of impacts, their mechanisms and solutions across both mental health and climate actions. We outline why these have

been identified as priorities and how they can be addressed by combining expertise across disciplines and sectors.

Research priorities are presented within four overarching research categories that were identified as areas of critical need for further work globally and that map the climate and mental health research space at a high level, based on an initial review of global literature. Note that some priorities span multiple categories.

- **Impacts, risks and vulnerable groups:** improving our understanding of the extent to which mental health is affected by climate change and for whom. For example: what mental health outcomes are impacted or at risk; the prevalence, severity, economic and societal costs of these impacts; and who is most vulnerable to these impacts.
- **Pathways and mechanisms:** improving our understanding of how climate change affects mental health and whether there are factors specific to climate change that increase mental health risks or create new experiences of mental health challenges. This includes considering biological, psychological, societal or environmental pathways and mechanisms.
- **Mental health benefits of climate action (adaptation and mitigation):** understanding and quantifying when and how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health.
- **Mental health interventions/solutions in the context of climate change:** identifying the most effective mental health interventions/solutions/actions to support mental health in the context of climate change, across diverse sectors. This encompasses providing support to people already experiencing negative mental health impacts and reducing risk or severity of future negative impacts, as well as building capacity for addressing and implementing these mental health interventions.

We hope that these priorities will act as an inspiration to guide the direction of the research community of practice, provide investment targets for funders and focus generation of evidence to best enable policy makers and practitioners to address the emerging and predicted mental health needs in response to climate change in CSA.

Priority research themes

In the priority research themes table below, we use mental health challenges as they relate to established mental health issues and disorders; these include depression, panic disorders, anxiety, stress, PTSD, paranoia, substance misuse, etc. Communities within the CSA region mostly use 'mental wellbeing' language to describe variations in emotional states; these may or may not constitute mental health challenges. However, using mental wellbeing language is more inclusive of communities in the region and would enable research to better connect with issues at the ground level.

Table 5: Priority research themes for CSA

Research category	Priority research theme
A. Impacts, risks and vulnerable groups	1. Understanding the climate-specific mental health risks and challenges for vulnerable groups (such as farmers, the elderly, adolescents, informal urban workers, Indigenous communities, women and children) across CSA. 2. Understanding the long-term effects on mental health resulting from the indirect impacts of climate change, such as food insecurity, loss of livelihood, damage to housing, etc.

- Quantifying the additional mental health burden related to food insecurity attributable to climate change.
 - Quantifying the additional mental health burden related to loss of livelihood due to climate change.
 - Quantifying the additional mental health burden related to loss/damage to housing due to climate change.
3. Assessing the long-term direct and indirect mental health impacts of extreme climate events and climate disasters (extreme heat and drought, cyclones, flooding, etc.).
 4. Understanding how responding to climate-related disasters impacts the mental health of caregivers and service providers during and after disasters, and the key factors contributing to this.
 5. Understanding how frequent glacial flooding affects Indigenous communities (e.g., effects on infrastructure, education, and livelihood) in the Himalayan region and the likelihood of experiencing mental health challenges related to these effects. This includes identifying specific groups that are most affected. *(Note: this would likely need to be a qualitative study given the scarce population in the Himalayan region.)*
 6. Identifying which mental health challenges are increasing in frequency or severity in the region as a result of climate change, particularly considering challenges beyond PTSD, stress, anxiety and depression. These may include alcohol abuse, substance misuse, self-harm and suicidal ideation.^{26, 27}
 7. Understanding the specific mental health risks faced by the most vulnerable groups of climate-induced migrants (e.g., persons with disabilities, the elderly, adolescents, pregnant and lactating women, informal urban workers, children and gender-diverse communities) during and after climate-related displacement.
 8. Exploring how climate change intersects with gender-based violence and understanding the related mental health impacts, particularly on vulnerable populations across age groups.
 9. Measuring the extent to which the impact of heat on ecosystems, livelihoods and overall mental wellbeing in the region leads to an increased prevalence of stress, anxiety and depression. With respect to farmers in the region, understanding livelihood impacts on the increased rates of suicides.
 10. Understanding how reproductive health problems related to the effects of climate change (such as salinity or climate disasters) affect the mental wellbeing of women and may increase the risk of mental health challenges. *(Research should cut across socioeconomic strata and include those living in rural, semi-urban and urban poor localities).*
 11. Understanding how climate disasters and slow-onset climate impacts like increased temperatures affect children's risk of mental health challenges.
 12. Understanding the climate-related anxieties of coastal populations and their impacts on mental health challenges.
 13. Improving the psychometric measures of climate-specific mental health

and wellbeing outcomes resulting from climate-induced disasters. For example, exploring ways to incorporate lived experiences into these assessments. *(In addition to quantitative surveys, qualitative phenomenological methods²⁸ would provide an understanding of the lived experience of a phenomenon [climate disaster] at the individual level as well as the social contexts in which they are embedded.²⁹ This would assist in the development of context-specific interventions and the appropriate application of mental health measures.)*

14. Understanding how different factors (community resilience, social support, government and NGO support) shape community perceptions of climate change-induced mental health challenges, including the perceived degree of urgency to address these challenges.

B. Pathways and mechanisms

1. Designing a comprehensive, effective, and culturally sensitive biopsychosocial model to conceptualise and assess the mechanisms of climate-induced mental health challenges in the region. *(This should incorporate biological, psychological, socioeconomic, socio-political, cultural, ecological, and environmental determinants.)*

2. Working with governments, civil society, NGOs, volunteer organisations and research organisations to create an inclusive framework for developing interventions for climate-related mental health challenges. *(This framework should incorporate perspectives of those directly impacted by climate change, such as farmers, and experts in the field, such as climate scientists/researchers.)*

3. Understanding how cumulative exposures to multiple climate-related stressors contribute to deteriorating mental health among vulnerable groups.

4. Understanding the extent to which awareness or lack of information/misinformation/uncertainty about climate change/climate hazards affects the mental health of affected individuals and vulnerable groups (e.g., frontline workers, Indigenous populations).

5. Assessing the extent to which climate change-related financial losses (loss of livelihood, infrastructure damage) contribute to or worsen mental health challenges in vulnerable populations (farmers, locals in coastal regions and people in flood-prone areas) in the region.

6. Exploring the relationship between climate change-related air pollution exposure and the risk of developing depression, anxiety, psychosis or neurocognitive disorders such as dementia across age groups in CSA.

7. Exploring the pathways and mechanisms by which extreme temperatures contribute to increased symptoms of mental health challenges (such as anxiety and depression) and mental wellbeing challenges (such as solastalgia and grief) in CSA.

8. Understanding how the immediate social impacts arising from climate disasters (e.g., loss of social capital, disruption in education, gender-based violence) contribute to psychosocial distress and affect mental health in affected communities.

9. Assessing the impact of climate-induced vector-borne and infectious disease spread on the mental health of affected individuals and their families.

10. Understanding how different communities perceive climate change-induced mental health challenges, including the perceived degree of urgency to address these challenges.

C. Mental health benefits of climate action (adaptation and mitigation)

1. Identifying and evaluating the climate change adaptation strategies/interventions (e.g., new agricultural practices, water management, infrastructure) in CSA and their benefits (direct or indirect) on mental health.
2. Identifying actions taken by local and national governments to address the impacts of climate change and climate hazards and evaluating their effects on the mental health of individuals affected by climate hazards.
3. Assessing the impact of community-based adaptation strategies (e.g., social strategies utilising peers, building connections between health workers and the community) on the mental health and mental wellbeing of vulnerable populations.
4. Assessing the variation in perceptions and understandings of climate change and climate hazards across diverse regions in CSA, taking into account local knowledge, cultural perspectives and regional contexts. This includes determining which factors contribute to these variations and how a nuanced understanding of these differences might inform region-specific climate adaptation strategies that benefit mental health.
5. Identifying the mental health benefits of actions that protect farmer livelihoods from prolonged periods of drought, salinity intrusion and flooding.
6. Assessing the mental health benefits of actions that reduce air pollution in CSA as part of climate mitigation initiatives, such as reducing the burning of fossil fuels.

D. Mental health interventions/solutions in the context of climate change

1. Outlining effective, locally relevant coping strategies to address mental health challenges induced by climate change and exploring how to integrate these strategies with established methods of psychosocial treatment.
2. Identifying the government support schemes (at all levels) across the region to address mental health concerns/distress during and after climate disasters or climate extremes (e.g., heat wave/cold waves).
3. Understanding the local terminology and definitions used to discuss mental health challenges and mental wellbeing issues and how they influence discussions about mental health in the context of climate change. This includes exploring how this terminology can be incorporated into new and existing interventions.
4. Evaluating strategies to ensure the widespread availability and accessibility of mental health interventions (including identifying stakeholders to implement and scale up these strategies).
5. Identifying and evaluating strategies to raise awareness of the mental health challenges of climate change among affected individuals to enable them to identify when and how to seek necessary support (e.g., by understanding the causes and symptoms associated with these experiences), and among policymakers to enable them to design and advocate for effective policies.

6. Identifying and evaluating training and interventions (through different sectors: government, non-government, research institutions) to equip community health workers to deliver frontline climate-related mental healthcare services.
7. Identifying methods to incorporate the cultural, contextual and local perspectives of on-the-ground mental health professionals when developing interventions for climate-related mental health challenges.
8. Understanding how cultural, social and economic characteristics influence perceptions and responses to climate change and mental health in CSA and how these factors should be considered when developing mental health interventions.
9. Exploring strategies to strengthen community-based initiatives and support systems to make communities more resilient to and responsive in supporting climate-specific mental health challenges.
10. Understanding how health systems and services can be made more responsive to climate-induced mental health needs within the community.
11. Determining the most effective short- and long-term preventive, promotive and rehabilitative interventions to address climate-specific mental health challenges.
12. Understanding the roles of an individual's support network (e.g., friends, family members), community and external resources (e.g., professional counselling, online resources) in improving outcomes for people experiencing mental health challenges following climate disasters and exploring how these can be strengthened.
13. Exploring strategies to provide training and interventions to equip community health workers to deliver frontline healthcare services to individuals affected by climate hazards and assessing the mental health benefits of these services.

Overview of themes for impacts, risks and vulnerable groups

Why were these themes chosen as priorities?

There is a need to understand the different impacts on urban, semi-urban and rural areas. Participants specifically mentioned the importance of including neglected vulnerable groups, such as farmers, the elderly, adolescents, and persons with disabilities. They reiterated the importance of understanding the risks faced by these groups before and after climate-induced displacement (Theme A7).

The most commonly raised climate disasters included cyclones, extreme heat, floods and landslides. Participants raised concerns about the perceived intensity of resulting mental health impacts from issues of livelihood loss, food security and infrastructure damage. The long-term effects of livelihood damage are areas that need to be further explored (Theme A2) (Theme A3). Furthermore, given the increasing frequency and severity of climate disasters, dialogue participants also emphasised the importance of understanding mental health challenges beyond PTSD, stress, anxiety and depression (Theme A6).

Specifically, the intersectionality of climate and gender, which includes reproductive health and gender-based violence problems, was considered very important (Theme A8), as was the emotional wellbeing of women given climate issues such as increased salinity in water (Theme A10).

Throughout the dialogues, participants reported a need for more research on the impact of heat on anxiety and depression. Very little is understood about the impacts on brain functioning and wellbeing (Theme A9). Particularly vulnerable groups to the effects of prolonged heat exposure included urban and rural outdoor workers, the elderly, pregnant women and children under the age of five (Theme A11).

There was particular emphasis on caregivers and the effects on their mental health following climate disasters; this group includes nurses and community workers, who are often tasked providing mental health interventions (Theme A4). Respondents also reported the importance of Indigenous communities, citing the communities of the Himalayan region in Nepal as particularly vulnerable group to glacial flooding (Theme A5).

What are examples of methodologies, metrics and datasets that could be used to address these?

Respondents suggested a combination of quantitative and qualitative methodologies to address these themes. Qualitative methods (e.g., community-based participatory research, ethnographic case studies) provide a means to understand the context of climate-related anxieties and mental health issues of smaller, hard-to-reach communities.

For example, it was suggested to utilise qualitative methods to understand the unique vulnerabilities where there are vulnerable groups in remote locations (e.g., Himalayas in Nepal). These methods are also useful where large-scale surveys cannot be implemented. Another suggestion for building data sets was to consult local mental health professionals, as they will be able to assist in identifying vulnerable groups and provide support for the research.

Overview of themes for pathways and mechanisms

Why were these themes chosen as priorities?

The effect of heat on mental health is not fully understood within the region, and there is a need to understand the mechanisms by which heat may affect mental health, including the impact of heat on brain function (Theme B8). Suggested areas of focus for understanding the pathways by which climate change affects mental health in the region include exploring the biopsychosocial elements of mental health issues (Theme B1). This is because impacts of climate-induced disasters affect many aspects of biology, psychology and the social environment, including cross-cutting impacts on infrastructure and links to air pollution (Theme B4).

“Also, mental health services are very much like a medical model. Here, for climate change, we really need a biopsychosocial model, a social model, a comprehensive model. Therefore, we have to think about these” (Participant, Dialogue 1)

With the high prevalence of pollution in CSA, especially urban cities, there is a double risk of heavy air pollution and climate induced disasters. A dialogue participant referred to studies^{30, 31} that have reported that particulate matter with an aerodynamic diameter greater than 2.5 µm has adverse effects on the functioning of neurotransmitters, leading to symptoms of anxiety and depression; studies have shown associations with anxiety,³² depression³³ and psychological distress.³⁴ However, similar work hasn't been conducted in CSA to understand the ways people may be similarly affected in the region (Theme B7).

In addition to understanding the overall effects of the environment on health, dialogue participants emphasised the importance of incorporating lived experiences of those suffering from mental health issues, in addition to using standardised psychometric measures (Theme B2). Lived experiences are also important to consider as they can uncover perceptions, lack of awareness on mental health issues and

livelihood challenges faced by those with mental health challenges (Theme B5) (Theme B12). These are a result of social and financial losses of different populations (farmers, vulnerable coastal populations) (Theme B6) (Theme B9).

Climate disasters also affect the rate and transmission of vector borne infectious diseases; this is an area that requires further investigation in CSA (Theme B11).

All dialogue participants agreed that there should be integration between governments, civil society and research organisations to enable meaningful climate action at the intervention and government level (Theme B3).

What are examples of methodologies, metrics and datasets that could be used to address these?

Respondents provided myriad suggestions on how to collect relevant data. One respondent suggested creating a framework that incorporates insights from local researchers in the region/area being studied. The framework would incorporate key variables for the region, identified through discussions between NGO workers, researchers and lived experience individuals. Others suggested using qualitative studies to identify the most vulnerable areas/populations; these include qualitative observations over different periods of time. Qualitative observations allow for an in-depth understanding of lived experiences and their context; also the flexibility of the method can help to overcome sensitivities associated with mental health research.

One dialogue respondent also suggested short term/small quantitative surveys over regular intervals, which would focus on key economic and livelihood related variables (household assets, income, job opportunities, reported nutritional intake). Another participant suggested developing DALYs specific to climate-mental health impacts.

Overview of themes for mental health benefits of climate action (adaptation and mitigation)

Why were these themes chosen as priorities?

There should be opportunities for all stakeholders to be involved in strategies or interventions for climate-related mental health issues, especially local communities (Theme C1) (Theme C2). The dialogues presented examples of locally led solutions for climate issues, including building community-financed small dams in Afghanistan and raising housing in Bangladesh.

Respondents reported the value of integration of local community-based climate action strategies and mental health initiatives (Theme C3). Therefore, research questions are based on the utilisation of local strategies, as well as the improvement of successful local government initiatives through further training and interventions (Theme C4).

It is also important to understand the effects of community-level strategies on individuals, as well as the associated perceptions. Therefore, questions prioritised not only local strategies but also sought to assess the perceptions and local understanding of climate hazards and mental health (Theme C4). As it was agreed that farmers are a particularly vulnerable group, participants highlighted the importance of identifying mental health interventions that could protect farmers during prolonged periods of drought, flooding, or salinity intrusion (Theme C5).

The impact of air pollution (across countries, regions, populations, and vulnerable groups) and its effect on climate-induced mental health issues was again emphasised. For example, there are differing impacts of air pollution and heat on urban populations (heat islands) as compared to those who are nomadic or live in rural areas. Therefore, it is crucial to understand the mental health benefits of actions that work towards reducing levels of air pollution (Theme C6).

What are examples of methodologies, metrics and datasets that could be used to address these?

Dialogue respondents did not provide any specific suggestions with regards to data sets or metrics to address the research questions. However, some suggested the use of climate scenario modelling in order to predict future challenges. This would assist with developing interventions in communities and understanding what would be required to address respective climate-induced issues in the region.

Furthermore, participants proposed the possibility of using legacy data from interventions for prospective research. An example of such an intervention is the pilot testing of Mental Health Gap Action Programme (mhGAP) in Pakistan.

“The Ministry of Planning & Special Initiatives has been testing a pilot project for Pakistan that mental health is going to be integrated in primary healthcare through frontline workers and primary healthcare physicians. Which means that they're going to use mhGAP. The pilot was started during COVID-19. It is now being tested in one of the most vulnerable provinces of Pakistan, Khyber, and Pakhtunkhwa, with populations that have been affected by climate change and also violence.” (Participant, Dialogue 2)

Using existing data from these types of government interventions would help answer research questions around the efficacy of government interventions, including what is required to improve or supplement them.

Overview of themes for mental health interventions/solutions in the context of climate change

Why were these themes chosen as priorities?

The majority of our themes/questions centre on utilising community-based support strategies along with interventions to understand mental health at the community level; whether new or existing interventions, either government or NGO (Theme D2). Dialogue participants particularly emphasised the importance of communities and professionals' understanding of the concept of mental health (Theme D1). Furthermore, understanding local mental health terminology can help overcome communication issues at the ground-level, leading to more effective intervention outcomes (Theme D3).

As with many communities in CSA, the concept of 'mental health' is often misunderstood or not acknowledged at all. As a result, it is especially important to understand local terminologies and definitions around mental health, as well as the perceptions of around climate and mental health (Theme D3) (Theme D8). Furthermore, it was suggested to incorporate on-the-ground professionals, such as frontline workers and community health workers, within design and implementation of interventions (Theme D7). These groups are able to provide information relating to cultural and social nuances of the communities in which they work (Theme D8).

Incorporating community members also presents an important opportunity to understand support networks (Theme D10), as well as understand cultural, social, and economic characteristics that influence perceptions of climate change and mental health (Theme D9). As a result, studies can better outline local coping strategies and their effectiveness compared with more established methods, enabling them to better inform health systems and (Theme D5) (Theme D11) (Theme D12).

An example of interventions integrating with Indigenous psychosocial knowledge was reported from Pakistan:

“NGOs have been really active in Pakistan for mental health. Recently, I worked for GiZ. And in the first phase of this project, they built the capacity of around 400 to 500 people in the community on basic counselling skills and psychosocial support for community and

one-to-one psychosocial support. Which meant that these people who are living in that community, they would be able to support their own community through Indigenous solutions.” (Participant, Dialogue 1)

Building on such interventions, dialogue participants mentioned the importance of raising awareness of climate-induced mental health and wellbeing challenges amongst individuals and policymakers alike (Theme D5). There were conversations on strategies that could be implemented to ensure widespread applicability of mental health interventions and the stakeholders required to implement them (Theme D6). Evaluating awareness raising strategies for affected individuals is required, as it would help them understand when and from whom to seek support (Theme D4). This would involve understanding an individual's support networks and the available external resources (online resources, professional counsellors) (Theme D12).

One of the dialogue participants stated that there would never be enough psychiatrists and psychologists to meet the needs of a country. Therefore, he suggested that improving the training of community health workers could work towards solving this gap.

“In our work, we’ve realised that we are never going to reach a point where they're going to be adequate psychiatrists and psychologists in the country to address problems like anxiety, depression, psychosis, etc. Our way of working around that problem is that you train a local cadre of health workers who do not have any previous specialist training.” (Participant, Dialogue 1)

However, further investigation is required to understand the feasibility of using community health workers for both research and implementation to address climate induced mental health issues in communities (Theme D13).

What are examples of methodologies, metrics and datasets that could be used to address these?

Frontline and community workers can contribute to the development and translation of key metrics, given their knowledge of their respective communities. One participant suggested that short surveys conducted at regular intervals would provide a better understanding of the situation. Data that could be collected includes socio-demographic information and the availability and reach of community health workers.

While no specific datasets or metrics were referred to, participants suggested more local research initiatives and funding for research. Technical knowledge needs to be passed on to the Global South where it is currently lacking. This is often the result of many skilled citizens leaving their home countries, as was reflected by one of the participants from Bhutan.

“I cannot see any kind of research studies done in Bhutan by the Bhutanese people. Many of the research that is available right now from my country is done by the people living outside of the country, this includes scientists and advocates. I think Bhutan would be able to do it far better, but there has been limited research and limited Bhutanese researchers. While there are numerous Bhutanese resources on ResearchGate, they are unavailable on prominent websites such as Google Scholar for example. I personally believe, this is due to lack of awareness of the platform for publishing research.” (Participant, Dialogue 1)

Action agenda

As the climate change and mental health field in CSA builds and the evidence base grows, it is crucial to avoid perpetuating existing challenges, including: 1) disconnections across disciplines and between researchers and policy makers; 2) unequal focus on topics and geographies; and 3) siloed decision making

for climate and for mental health. This action agenda sets out a shared vision as a rallying focus for the mental health and climate change field in CSA. This includes guidance on how to best support the growing community of practice, how to translate evidence to action and the principles that should guide this approach. Enacting this agenda will require transdisciplinary effort and coordinated action across research, research funding, policy and practice. This action agenda aims to guide this work by setting out the challenges which must be addressed, opportunities that can be harnessed and priority actions to work towards a thriving climate and mental health field.

Regional vision for mental health in a changing climate

Vision statement

CSA envisions a future as a leader of a climate-conscious revolution, where a 'healthy climate mind'ⁱⁱⁱ drives transformative action for mental health. Achieving carbon neutrality, resilient infrastructure and global collaboration, we cultivate a green economy that creates jobs and significantly cuts carbon emissions. Picture a 42-year-old father holding his once drought-burdened 12-year-old son's hand, proudly showing him a lush, green land—a testament to international support for climate change initiatives. Our vision encompasses social equity, inclusivity and proactive mental health preparedness. Our future thrives on exponential growth in inclusive spaces, a platform for identifying the mental health impacts of climate change and empowering vulnerable groups in climate response. With rational pasture use, resilient communities and environmentally sustainable practices, we minimise the impact of climate change, protecting mental health and providing mental health support for affected communities.

Key characteristics of the desired future state for climate change and mental health in the region

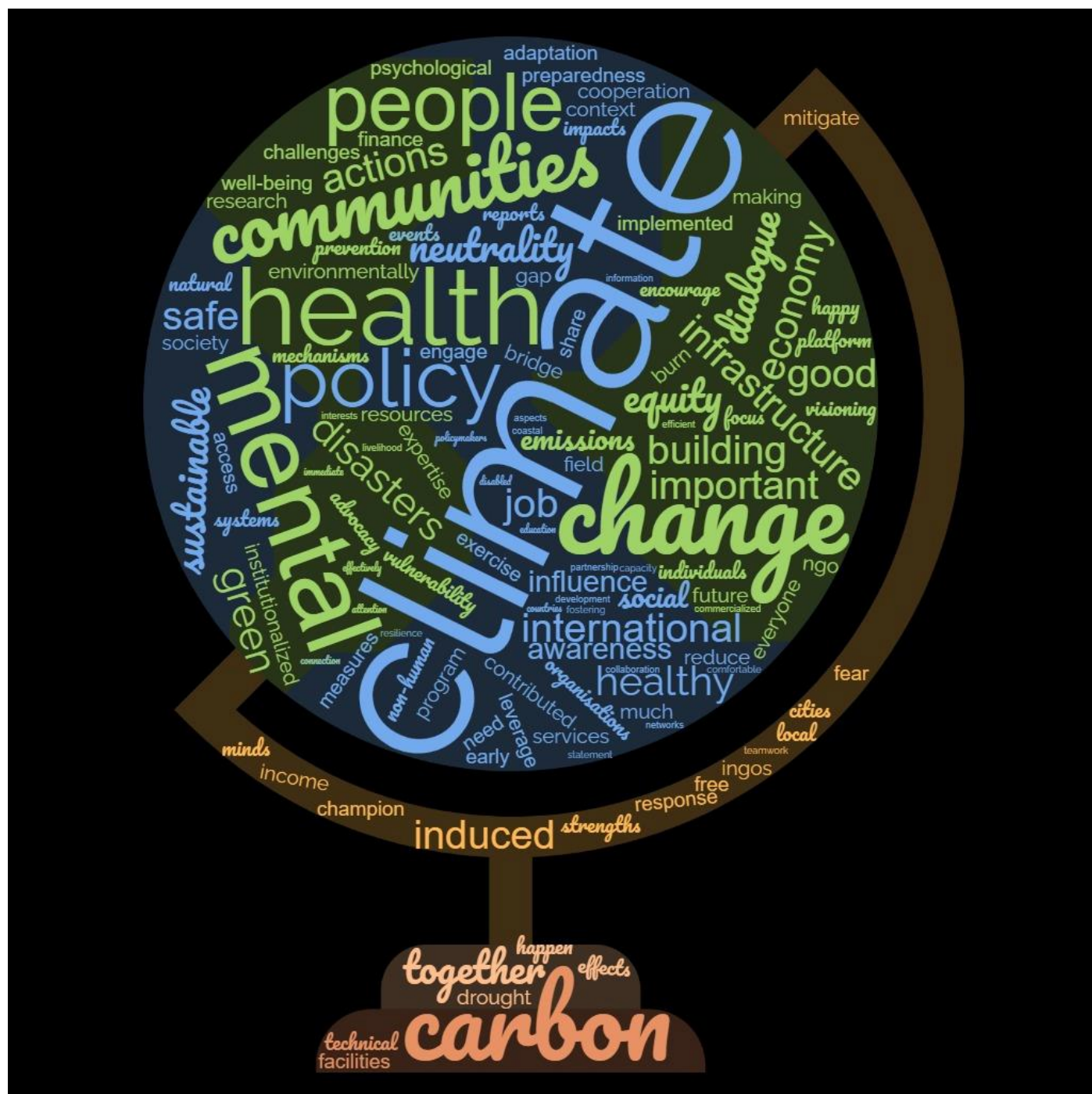
In 2030, countries in the CSA region will see resilient communities that have been able to successfully integrate mental health considerations into climate action initiatives. There will be widespread awareness, education, and support networks to help address the mental health impacts of climate change. Communities will be able to openly discuss and prioritise mental health, creating a culture of understanding and resilience. There will be institutionalised mechanisms that regularly report on the mental health status of the people, guiding evidence-based policy measures.

By 2030, the climate change and mental health research field will have flourished, with communities, researchers and policymakers equally committed to impactful projects in this nexus. Research outcomes will effectively inform policy decisions and offer practical solutions for vulnerable communities, particularly those with unique challenges (like persons with disabilities). Collaborative efforts of local and international organisations will contribute to a global body of knowledge.

A dynamic community of practice will be in place where members actively champion partnerships, collaboration and policy influence. There will be a shared commitment for advocacy for greater attention to mental health issues, particularly for those communities whose voices are often unheard. This community will stand united, resilient and prepared for future challenges, reflecting a shared vision of sustainable wellbeing.

Figure 2: Word cloud from visioning statement

ⁱⁱⁱ This concept encompasses recognising the interconnectedness of mental health and environmental factors, understanding the mental health implications of climate change and actively promoting community support, coping strategies and practices that lead to ecological and psychological wellbeing.



Creating an enabling environment for research at the intersection of climate change and mental health

The identified research priorities will only be of value if they are enacted. The climate and mental health field is relatively new and rapidly growing, and now is the time to ensure that it is designed to deliver a mentally healthier future in the context of the climate crisis. In a field that spans multiple disciplines and sectors, each with different cultures and ways of working, and on a topic with low awareness in many countries, what is needed to support capacity-building efforts? What principles must guide the field, and what are the challenges and opportunities in the region to create an environment that would enable such research?

This section presents a synthesis of dialogue discussions and inputs from co-conveners, the LEAG, YAs and other members of the community of practice on what is needed to implement the research agenda and foster an enabling research environment for climate change and mental health.

A desired state of climate change and mental health research in CSA

- **Modelling for future trends in climate change and mental health:** ‘Good’ research on climate change and mental health in CSA means not only understanding the current landscape but also modelling for future trends based on changing variables of climate and health indicators. Effective research should explore how changing climatic conditions impact mental health and model future scenarios. Research should generate knowledge and insights into the impact of climate change on mental health and wellbeing across different countries in the region so that policymakers can make informed decisions at the regional and national levels.

“Research should not only understand the present scenarios but the present mental health problems also that people face. It should also try and understand what the scenarios would be used in the future, because of the changing variables.” (Participant, Dialogue 2)

- **Prioritising vulnerable groups and understanding lived experiences:** Researchers need to prioritise vulnerable groups and capture lived experiences of people and communities whose mental health is directly affected by the climate crisis. A needs assessment should be conducted on vulnerable populations as a starting point to ensure that research efforts are tailored to address the specific challenges faced by these populations.
- **Ensuring inclusivity:** Research efforts need to prioritise inclusivity, ensuring that comprehensive datasets are produced that address the needs of marginalised groups, like Indigenous Peoples, whose voices are often not heard.
- **Fostering collaborations at sub-national levels:** Institutionalisation of surveillance systems and collaboration with government departments at sub-national levels can provide a comprehensive picture of the challenges posed by climate change on mental health in the region.
- **Promoting local knowledge:** Qualitative methodologies, anthropological and behavioural science approaches can be used to understand and promote local knowledge to empower communities. A holistic understanding of the impact of environmental stress on human psychology can be understood using community-based participatory research methods through regular dialogues with community representatives, input from medical anthropologists, researchers and implementing partners to help unpack these concepts.
- **Communicating using familiar mental health terminologies:** Knowledge sharing, accessibility and the use of bottom-up approaches to generate evidence are important. Effective communication using well-defined, familiar mental health terminologies is important to ensure research findings are translated into language that is accessible for general understanding.

Challenges holding back research

- **Social taboo and stigma around mental health:** Whether in Bangladesh, India, Nepal or Afghanistan, there’s a prevailing reluctance to openly discuss mental health, hindering progress in research.
- **Gender restrictions:** Gender-related restrictions in Afghanistan act as a barrier, limiting direct research participation for women and children, who are some of the most vulnerable groups for climate change and mental health.

- **Lack of funding for research:** Government ministries and relevant authorities often allocate limited funding to research the impact of climate change on mental health due to lack of awareness on the urgency and significance of their intersection. Further, collaborative research efforts with international partners may face challenges in securing funds. Climate change and mental health research often involves interdisciplinary approaches and long-term studies, making it resource-intensive, making funders hesitant to invest.
- **Language barriers and varied cultural understandings of mental health:** Mental health discussions, particularly those related to climate-induced distress, involve culturally sensitive topics. Expressing these concepts in a manner that is culturally appropriate and does not stigmatise mental health issues can be challenging when language nuances are not fully understood.

"One of the challenges would be the language that we use to discuss with the community because sometimes we don't use the same terminology as them in the community. We need to understand very clearly that what we are asking them is the same meaning to them as it is to us. Selection of proper words and language is very important. And so that's a challenge. We need to learn from the community how they express their understanding of what's going on. It may not be the same as us, as researchers." (Participant, Dialogue 2)

- **Mismatch of priorities between communities and researchers:** Communities often have immediate concerns and priorities that may differ from research agendas set by academic or scientific institutions. For example, communities might prioritise access to economic resources, clean water or infrastructure development over mental health concerns related to climate change. Researchers should work with local communities to set the research agenda and identify key questions that are relevant and beneficial to all groups.

"Another challenge is that whether we can align what their (community) needs are with the needs of the research questions. Sometimes they want us to study different things that we might not be capable of or help them with those issues." (Participant, Dialogue 2)

- **Lack of local representation in research:** In Bhutan, for example, the lack of local researchers currently contributing first hand and limited education opportunities within the country, especially beyond the bachelor's level, contribute to a shortage of local research expertise.

Opportunities and enablers

- **Limited existing research on mental health and climate change:** Limited existing research presents a significant and largely untapped opportunity for further exploration and understanding, using innovative methodologies, identifying new connections and developing targeted interventions. This research gap also provides the opportunity for publishing research in reputable journals to help disseminate findings and raise awareness among relevant stakeholders.
- **Fostering meaningful engagement with communities to coproduce knowledge:** Incorporating a community-based participatory research approach is essential for meaningful engagement with communities. This strategy involves communities from the inception of research, ensuring their needs and perspectives are considered. It is an opportunity to build trust, enhance relevance, provide an opportunity for the co-creation of interventions, and increase the impact of research outcomes. Using this approach can encourage individuals with lived experience to participate in research using a bottom-up approach. This will not only generate reliable evidence but also promote inclusivity and cultural relevance of research findings. It is also important to recognise the willingness of youth to address climate change in their regions and to employ them as agents of

change. Involving youth in research initiatives and advocacy efforts can help them identify their commitment to effective climate action.

"We should encourage people with lived experience to write or take part in research more actively, possibly through a bottom-up approach to generate evidence." (Participant, Dialogue 2)

- **Collaboration across sectors:** It is important to identify and engage with stakeholders at sub-national, national and regional levels. One example of such collaboration is the opportunity to collaborate with community development councils in Afghanistan to help reach women and children as research participants. Collaborating with mental health institutions and organisations can facilitate the integration of climate change-related questions into mental health assessments. This would allow for a deeper understanding of the impact of climate change on mental health, providing valuable data for research and policymaking. Further, collaborating with government bodies, establishing coordination with ministries and building global partnerships can facilitate research and action in the region.

"An opportunity is the availability of CDCs. CDCs are available across the country in all 34 provinces of Afghanistan. These CDCs can help us access vulnerable groups like women and children" (Participant, Dialogue 2)

Relevant potential partners

The following table outlines relevant potential partners categorised by stakeholder type in CSA.

Table 6: Regional stakeholders and potential partners for research in CSA

Stakeholder type	Primary stakeholders*	Secondary stakeholders**
Government bodies (ministries and local government bodies)	● Bangladesh Bureau of Statistics ● Ministries of agriculture, irrigation and livestock ● Ministries of environment, forests, and climate change ● Ministries of health and family welfare ● Ministries of public health ● National Environmental Protection Agency, Islamic Republic of Afghanistan	● Local government departments ● Ministries of education ● Ministries of water resources and agriculture ● Processing Industry of the Kyrgyz Republic
International NGOs	● Action Aid ● CARE ● Plan International ● Save the Children	● Air Quality Central Asia Dialogue Platform ● International Fund for Saving the Aral Sea (IFAS) ● The Regional Environmental Centre for Central Asia (CAREC)
National & grassroots NGOs	● Friendship ● Green Alliance KG ● SAJIDA Foundation ● Unison Group	

Research organisations	● Afghanistan Research and Evaluation Unit ● BCAS, Bangladesh ● C3ER, BRAC University, Bangladesh ● Centre for Chronic Disease Control ● Centre for Health analytics Research and Trends (CHART), Ashoka University ● ICCAD, Bangladesh ● Institute for Behavioural Health and Applied Research AUCA, Kyrgyzstan ● International Institute for Central Asian Studies ● Natural Resources Defence Council ● Tian Shan Policy Centre (TSPC)	
UN organisations	● Food and Agriculture Organization ● UNDP ● UNESCO ● UNICEF ● World Food Programme ● World Health Organization	
Funding agencies	● FCDO ● Green Finance Centre Bishkek ● IDRC ● U.S. Embassy in the Kyrgyz republic ● USAID ● Wellcome Trust	
Communities	● Organisations/community-based organisations – e.g., community development councils (in Afghanistan)	● Communities engaged in climate action at grassroots level ● Communities engaged in mental health actions ● Communities vulnerable to impact of climate change ● Local leaders ● Youths from affected communities
Mental health stakeholder groups	Mental health researchers	● Mental health activists ● Mental health experts/professionals ● Psychiatrists ● Psychologists
Others stakeholder groups	Medical colleges	● Civil society organisations ● Influential world leaders ● Pharmaceutical companies ● Youth activists

*Primary stakeholders are directly involved in the research

**Secondary stakeholders are indirectly involved in research initiatives

Priority next steps/recommendations to investors and actors

Addressing the intersection of climate change and mental health in CSA requires comprehensive, collaborative efforts and a nuanced understanding of the region's unique contextual factors. Some priority next steps are listed below:

- **Foster a better understanding of the linkage between climate change and mental health.** A taboo surrounds the issue of mental health, compounded by a lack of climate literacy that hinders people from connecting these two issues. Recognising this, there is a pressing need for initiatives to foster better understanding, beginning at the community level, to facilitate the linkage between mental health and climate change. Such efforts will not only stimulate greater demand for evidence generation but also streamline the identification of direct and indirect impacts of climate change on mental health and overall wellbeing. Additionally, they will facilitate a deeper understanding of the nuances of these incidents and their regional implications.
Stakeholders: Research organisations, community health clinics, NGOs
- **Empower communities through training and awareness raising.** In addition to engaging influential community members in mental health and climate change learning sessions, there is a crucial need for sensitising local communities through training and awareness of community members. Organisations and government institutions should collaborate with public figures, influencers and civil society organisations to raise awareness about the mental health impacts of climate change. This collaborative effort should involve setting up learning sessions for selected personnel, including influential community members. Through these sessions, key messages regarding climate-related mental health issues can be discussed and effectively communicated. This comprehensive approach lays the groundwork for future community engagement, facilitating increased awareness and empowerment among local communities.
Stakeholders: Community leaders, government agencies, community health workers
- **Strengthen collaboration between climate experts, mental health researchers and social scientists through a series of research workshops.** To create a comprehensive understanding of the complex intersection of climate change and mental health, it is important to establish interdisciplinary research platforms and jointly work on research projects. Research should target practice through cross-disciplinary workshops, dialogues and training which can encourage knowledge exchange and bridge gaps in research and intervention. Research workshops could focus on a listing of priorities by different researchers, which can form the basis for funding initiatives for cross-disciplinary research. An example of this is addressing the burden of pollution with extreme heat on vulnerable populations (e.g., urban informal workers, and climate migrants); this would require expertise in environmental science, mental health, anthropology and statistics.
Stakeholders: Researchers and universities (social sciences, mental health, psychology, psychiatry, neuroscience), independent think tanks (health, policy)
- **Prioritise co-production of knowledge with affected individuals and communities and set up data-sharing protocols across different disciplines.** Policymakers and other relevant stakeholders (such as community members, health service workers, researchers, and University academics) should promote co-creation or co-production of knowledge. By valuing local knowledge and diverse perspectives, researchers can ensure that their work addresses the specific needs and concerns of communities. This can be achieved by establishing data-sharing protocols, where health service workers/organisations and researchers share data as it is being collected (or as soon as it is available). Examples of data sharing could be around health service visits, local perceptions as recorded by studies, and serious cases as they relate to mental health.
Stakeholders: Research organisations, universities, health clinics, hospitals, NGOs

- **Leverage research collaborations to ensure knowledge transfers to intervention initiatives.** The collaborations between research organisations and those implementing interventions should be strengthened. This may be achieved through integrating researchers within the formal decision-making process for interventions, creating a trust-based network which can facilitate appropriate knowledge transfer at the intervention level. Implementation would require stakeholder meetings/webinars prior to and during interventions; these serve as meetings to update implementers and NGOs to receive updates on the latest research.
Stakeholders: Research organisations, universities, NGOs, government agencies
- **Address the funding gaps in research by initiating dedicated funds at the national level.** Policymakers need to prioritise allocation of financial resources to interdisciplinary studies that looks at the intersection between climate change and mental health in CSA. A nationally dedicated fund can be established by government ministries and agencies to support research in this important intersection of mental health and climate change. At the same time, research organisations, local and international NGOs can work toward capacity building for researchers and practitioners. This will facilitate future funding initiatives.
Stakeholders: Local and international NGOs, government ministries, multilateral and bi-lateral agencies

Translating a growing evidence base into action that can respond to the mental health impacts of climate change

The current evidence base on the interconnections between climate change and mental health compels action in policy and practice to protect escalating mental health needs and promote co-beneficial climate actions. How can current evidence and new insights created through implementing this agenda best translate into changes to policy decisions and practices across both climate and mental health regional spaces?

This section presents a synthesis of dialogue discussions and inputs from co-conveners, the LEAG, YAs and other members of the community of practice, setting out the challenges and opportunities to translate evidence generated through research into policy and practice.

A desired state of climate change and mental health to action in CSA

- **Creating evidence-based policies underpinned by lived experiences:** Policies should be backed by evidence and lived experience stories, recognising, and acknowledging that power should be in the hands of those with practical knowledge to facilitate successful implementation. The discussions also brought to light the importance of funding for the Global South for impactful storytelling and research, emphasising the need for funding to be readily available, not solely through competitive calls. Effective action on climate change and mental health requires policy that is grounded in evidence, particularly from communities that are directly impacted by climate change in the region.
- **Incorporating voices of the most vulnerable and cultivating a sense of agency:** The power dynamics and hierarchies within vulnerable groups, communities, and different genders must also be considered in policy development. A bottom-up approach is necessary for effective policymaking, encouraging community involvement in policy development and implementation.
- **Multi-sector collaboration for multiple climate disasters:** Policy recommendations for climate action and mental health cannot follow a 'one-size-fits-all' approach and need to account for multiple climate disasters, regions and thematic areas. Policies should incorporate the voices of those directly affected, as well as involve experts in climate change and mental health to promote

interdepartmental and multi-sectoral coordination. Such levels of coordination can include research and implementation initiatives between university departments in environmental science, public health and physics with government meteorological departments, as well as commercial sectors like pharmaceuticals and readymade garments (addressing water and effluent treatment).

- **Realistic and feasible timeframe for policies:** Policies should have realistic and feasible timeframes for achieving objectives, with a focus on addressing mental health effects in specific sectors (like agriculture) and an equitable distribution of resources, technical know-how and power to foster meaningful change in CSA.

Challenges holding back translation of evidence

- **Lack of community awareness on the link between climate change and mental health:** The lack of awareness among communities about the linkage between climate change and mental health issues hinders the development of informed and resilient communities. For example, in Afghanistan, despite evidence of mental health impacts due to climate-related incidents like crop failure, local communities do not directly attribute mental health concerns to climate change. Their experiences of the war further complicate their perceptions of this interconnectedness.

"People do not directly relate mental health to climate change because in the last 40 years we have had war in Afghanistan, and people link the issue of mental health with war and the internal fight that we had...So yes, of course there are links for climate change and mental health. But at the local level, people do not think like that, even though there are evidence that people have anxiety when they lose their crops due to have flood" (Participant, Dialogue 2)

- **Poor coordination among relevant stakeholders:** National-level policies on climate change often fail to address mental health issues, resulting in a critical gap in necessary action plans. Due to this, there is also a lack of coordination between important actors for climate change and mental health. There is a need for better coordination between climate adaptation discourse and the global health community; bridging this gap is essential for more effective climate and mental health interventions. While policies, action plans and disaster response strategies exist for most countries in the region, they often lack accompanying implementation documents, or just exist as blueprints, delaying timely responses. This leads to a gap between policy formulation and effective action on the ground.

"I think the climate discourse from the climate change players has been evolving for 25 years, somewhat separately from the public health space. And I think our challenge has been to bring these communities of donors and actors together, which is not easy at the national level." (Participant, Dialogue 2)

- **Inequalities in research funding processes:** The inadequate transfer of technical know-how from the Global North limits the ability of the Global South to adapt to the climate crisis. The competitive nature of funding processes, favouring Global North applicants, perpetuates inequalities in research funding and limits the representation of Global South perspectives. Furthermore, limited prioritisation of funding for climate change and mental health initiatives, particularly in regions like Afghanistan, also hinders implementation of these projects. For the limited research projects that receive funding, challenges persist in ensuring that research dissemination efforts lead to meaningful policy integration, with a lack of follow-up on the long-term impact of findings on national policies. For example, in Bangladesh, the gap lies in the timeline required for research to influence policies, necessitating improved mechanisms for continuous monitoring and integration.

Opportunities and enablers

- **Absence of policies pertaining to mental health impacts of climate change:** The absence of relevant policies presents a prime opportunity for lobbying efforts emphasising the inclusion of mental health in existing climate change, as well as overall health, policies. For instance, in India, where there is widespread recognition of the impacts of climate change on mental health, focused efforts can ensure the effective translation of policies into actionable plans, bridging the implementation gap. In Bangladesh, there is an opportunity to lobby for integration of mental health issues into the country’s Health National Adaptation Plan (HNAP), which is yet to be finalised. In India, mental health has been included in its National Action Plan for Climate Change and Human Health. This plan has been adapted by states to develop customised state-level action plans for human health, which also include a focus on mental health.
- **Growing understanding of climate-mental health in the region:** There is opportunity to actively involve communities in climate and mental health initiatives, which will in turn help build resilience. Countries in the region can develop context-specific community engagement strategies, acknowledging the unique challenges each region faces, such as flooding or drought.

"There’s a growing understanding of mental health, generally, among the public. I mean, now, there are even celebrities championing mental health, so it’s become public discourse." (Participant, Dialogue 2)

- **Existing research on climate change:** Many of the countries in this region – for instance, Bangladesh – have existing research on the impacts of climate change, vulnerable regions and affected populations. This presents an opportunity to focus specifically on what’s missing, such as related mental health impacts, rather than starting from scratch.
- **Collaboration across sectors:** It is essential to build collaboration across different sectors (such as health, disaster management, education, etc.) for a more holistic approach in policymaking. Mental health initiatives need to be aligned with broader climate and health strategies. Collaborative efforts in India, for example, can leverage existing climate change and health policies to strengthen the integration of mental health considerations. In Afghanistan, collaborations with the Natural Disaster Department can facilitate a multi-sectoral approach to address mental health impacts linked to climate events. The growing focus on mental health issues in countries such as Bangladesh and India also offer an opportunity for mobilising resources, targeting capacity building efforts securing funding for implementation of relevant action plans.

"In a situation like climate change where it’s rapidly changing, where the impacts are rapidly changing and is overarching a lot of scope, it’s important we get the communities engaged, so the health systems will be supported and be more resilient." (Participant, Dialogue 2)

Addressing these challenges requires targeted actions, including policy advocacy, multi-sectoral collaboration, community engagement and resource mobilisation. By capitalising on these opportunities, we can make significant strides towards integrating climate and mental health actions in CSA.

Relevant potential partners

Addressing the mental health impacts of climate change requires a collaborative effort to translate evidence into actionable policies and practices. Engaging with key stakeholders is crucial for effective implementation. The following table outlines relevant potential partners categorised by stakeholder type in the CSA.

Table 7: Stakeholders and potential partners for policy and action in CSA

Stakeholder type	Primary Stakeholders*	Secondary Stakeholders**
Government bodies (ministries and local government bodies)	● Ministries of agriculture, irrigation and livestock ● Ministries of earth sciences ● Ministry of environment, forests and climate change/ministry of climate change, environment and energy ● Ministries of health and family welfare ● Ministries of housing and urban affairs ● Ministries of water resources, agriculture/ministry of Jal Shakti (water) ● Ministry of women and child development ● National environmental protection agencies	● Bangladesh Bureau of Statistics (BBS) ● Local government departments ● Ministry of education ● Ministry of public health ● Union Parishad (Bangladesh)
International NGOs	● CARE ● PATH International ● Plan International ● Save the Children ● WaterAid	
National and grassroots NGOs	● Air Quality Central Asia Dialogue Platform ● BRAC ● International Fund for Saving the Aral Sea (IFAS) ● Rupantar, Bangladesh ● Sangath, India ● The Regional Environmental Centre for Central Asia (CAREC) ● Uttaran, Bangladesh	● SAJIDA Foundation, Bangladesh ● U-Inspire Maldives

Research organisations		● BCAS, Bangladesh ● C3ER, BRAC University, Bangladesh ● Centre for Chronic Disease Control ● Centre for Health analytics Research and Trends (CHART), Ashoka University ● ICCAD, Bangladesh ● Natural Resources Defence Council
UN organisations	● Food and Agriculture Organization ● UNDP ● UNEP ● UNESCO ● UNICEF ● World Food Programme ● World Health Organization	
Funding agencies	● Asian Development Bank ● European Union ● FCDO ● Green Finance Centre Bishkek ● SIDA ● USAID ● US Embassy in Kyrgyzstan republic	
Communities	● Communities engaged in climate action at grassroots level ● Communities engaged in mental health actions ● Communities vulnerable to impact of climate change ● Local leaders ● Youths from affected communities	● Community development organisations/community-based organisations
Mental health stakeholder groups	● Mental health activists ● Mental health experts/professionals ● Psychiatrists ● Psychologists	● Mental health researchers

Others stakeholder groups	● Celebrities, influencers and public figures ● Centre for Mental Health, Law and Policy, Pune, India ● Civil society organisations ● Climate Parent Fellowship, India ● International Federation of Medical Students Association (regional/local chapters) ● Religious leaders ● Youth activists	
---------------------------	---	--

*Primary stakeholders are directly involved in policy and action

**Secondary stakeholders are indirectly involved in policy and action

Priority next steps/recommendations to investors and actors

Addressing the intersection of climate change and mental health in CSA requires comprehensive, collaborative efforts and a nuanced understanding of the region's unique contextual factors. Translating emerging evidence into impactful policies and practices requires collaboration and targeted efforts. Some priority next steps are listed below:

- **Incorporate climate-induced mental health issues in formal medical education.** It is crucial to incorporate the issue of climate change impacting mental health into mental health medical education and training curricula. The inclusion of this topic would help mental health professionals better understand and address the psychological impacts of climate-related events on individuals and communities. Workshops can be conducted with stakeholders to discuss syllabus gaps, compare medical school curricula across countries and plan for revisions. This should be followed by setting up appropriate advisory groups to begin implementation.
Stakeholders: Medical professionals, psychologists, psychiatrists, medical school regulatory bodies, relevant government ministries
- **Raise awareness of mental health services that address psychological effects of climate change in order to encourage affected individuals to seek care.** Raise awareness among the general population by highlighting the impact of climate change on mental health in community-level meetings and using media (TV, flyers & brochures, social media pages, websites) to disseminate messages on seeking mental health services.
Stakeholders: Communities (including marginalised groups), influential members (religious leaders, teachers, local politicians) NGOs, radio & TV stations, local government bodies)
- **Identify groups or organisations working on mental health resilience to identify areas for synergy and increase multisectoral and inter-ministerial collaboration.** Create a map of local organisations working on mental health, listing their target groups and competencies. This information can be used to address resource and funding gaps.
Stakeholders: NGOs and government agencies
- **Create more funding opportunities for programmes/initiatives that promote mental health resilience among communities affected by climate hazards.** The concern that people may not be aware of mental health services highlights the importance of raising awareness about the availability and accessibility of mental health services. The reason why individuals are less inclined to seek mental health services or support at lower levels of the healthcare system may be attributed to the

quality of care provided or the stigma associated with mental health, as well as the fear of neighbours and local communities becoming aware of one's mental health condition. These situations necessitate attention and the implementation of strategies to address and alleviate these barriers.

Stakeholders: Funders, NGOs, government agencies

- **Promote policy advocacy at the nexus of climate change and mental health.** Dialogues emphasising the importance of integrating mental health impacts into existing climate, health and education policies should be held between government ministries, local government bodies, donors, international NGOs, local organisations and vulnerable communities.

Discussion: strengths, limitations and next steps for the research and action agenda

Strengths

The strengths of this initiative are:

- **Raising awareness:** These dialogues helped raise awareness on the important nexus of climate change and mental health in the region. Many participants expressed that this was the first time such an important and up-and-coming topic was being discussed in this capacity with a large number of experts working in climate change and/or mental health across CSA; experts represented different fields, including research, implementation, advocacy and policy making.
- **Congregation of multiple stakeholders:** The dialogues gathered multiple stakeholders, who traditionally may have never crossed paths. This includes individuals with lived experience, either those who have experienced a mental health disorder or have experienced distress because of climate vulnerabilities.
- **Sharing of methodologies and interventions:** Overall, the research initiative proved to be a success. The strengths with respect to implementation were based on the combination of methodologies, understanding bottom-up approaches to interventions, sharing of knowledge and implementation techniques.
- **Incorporation of lived experiences:** This provided insight into lived experience stories and how they could be related to established research methods.
- **Variety of activities during the dialogues:** Multiple activities in the dialogues such as the visioning exercise, spectrum mapping and breakout discussions kept participants engaged, preventing any sort of 'Zoom fatigue'.
- **Sustained interaction before and after the dialogues:** Interactions with participants before, during and post-dialogue enabled the development of professional relationships, where research and other work was shared for the scoping exercise and dialogues.
- **Policy and implementation discussions:** The second dialogue was crucial for understanding the policy and implementation perspectives of the different countries; the majority advocated for evidenced based policies and integration of mental health into the existing country policies. Other suggestions included taking advantage of qualitative methodologies in remote areas and communication of initiatives through community leaders and celebrities.

Limitations

While these regional dialogues and engagements aimed to address the disconnected but deeply interconnected area of climate change and mental health, there were some challenges during the process. These include:

- **Uneven turnout of participants:** In CSA, the dialogues saw a greater turnout of climate experts than those in mental health, which meant that some of the discussions were heavily based on climate issues and agendas. The lower turnout of mental health professionals might be due to time management, as many psychiatrists, psychologists, psycho-social workers work during the afternoon (when dialogues took place).
- **Lack of equal country representation:** There was also a lack of equal representation from all countries in the region, particularly from countries in Central Asia. Besides this, one of the main challenges faced before the dialogues was the invitation and follow up process with participants, as some organisations were only able to send representatives.
- **Connectivity issues:** Connectivity issues across the region hampered conversations during the dialogue, especially in the breakout rooms. For some rooms, initial participants were unable to return for the second sessions.
- **Time constraints:** Time constraints limited conversations; given the breadth of experiences of participants, there was not enough time to explore go through all perspectives.
- **Low responses to the post-dialogue survey:** There was a low response rate for the post-dialogue survey. The length of the post-dialogue survey may have discouraged respondents from completing it; many respondents started the survey but did not finish. Despite follow-ups, other respondents were unable to commit time to the survey due their responsibilities at work.
- **Participant recruitment:** We encountered challenges in participant recruitment, employing methods like web searches and leveraging existing networks through a snowballing approach. This process introduced the possibility of biases, particularly participation bias, given the varied sources.

Despite these challenges, we diligently attempted to include participants representative of diverse countries, regions, sectors and expertise to enhance inclusivity. Acknowledging this limitation, we recognise it as a crucial initial step in establishing a network of key stakeholders. Sustained over time, this network holds the potential to evolve into a comprehensive database encompassing organisations and experts actively engaged in addressing challenges in the field of climate-mental health.

Conclusion

The intersection of climate change and mental health is an emerging area of research; efforts to understand this area in CSA have thus far been dispersed and limited. Through an innovative participatory approach, the CSA RCT organised two regional dialogues, which brought researchers, mental health professionals, individuals with lived experiences and practitioners behind one initiative. This enabled the region to produce an informative regional research and action agenda that highlights the most pressing issues and possible action points as they relate to climate change and mental health.

The project led to the development of an expansive and diverse RCoP, a first for the CSA region. The RCoP has brought together researchers, practitioners and individuals with lived experience from a diverse range of backgrounds, which include public health, psychiatry, sociology, disaster management, economics and psychology. The diversity of dialogue participants is reflected in the 43 priority research themes in this agenda.

To ensure continued collaboration of the RCoP and that the research themes are carried forward in the region, there is a pressing need for further investment in this area. Increased funding will strengthen the

RCoP, enabling this group to produce research that prioritises vulnerable and marginalised groups within the region, promoting representation of lived experiences at all levels. This research will in turn strengthen existing and emerging interventions to address climate-induced mental health issues.

Hearing from the Regional Community of Practice

Dialogue 1

- “The research on mental health is still less. We have not yet seen the relationship between mental health and climate adaptation or climate change. It is very important to explore how climate change is affecting mental health.”
- “We always say that this group is a marginal group or a vulnerable group. In the case of climate change, this vulnerable group or marginal group may be different. So we have to see who is affected the most and whose mental health is affected the most due to climate change. We need to first understand who the marginalised groups are, and then develop interventions specific to their needs.”
- “When there is a heatwave, there is a shortage of water. You don't get water. You go looking for water. You bring the water. You get late to work. You don't get to work. You don't get paid. You can't feed your kids. You can't take care of your family. It's a vicious cycle for people who have nothing much. This also causes mental health problems.”
- “In Sri Lanka, we have this unique problem where among the farming communities, the water is polluted. By using that water, lots of people have been affected with kidney disease. We are calling it chronic kidney disease. It's common in farming communities where, by 40, their kidneys are badly affected. That is attributed to water or maybe fertilisers that affect the water. This has become a very common problem. And with kidney diseases, there's quite a lot of suicides also happening. By their 40s, they have to have a transplant and they can't afford that. They eventually end up taking their lives.”
- “There's been research that shows rising temperatures have increased hospitalisations for psychiatric emergencies, increase in self-harming and suicidality. For people with psychiatric conditions such as bipolar disorders, these outcomes can be even more extreme.”

Dialogue 2

- “We are in a situation where the climate crisis is not going to go away, but we cannot continue with this level of unequal burdens anymore. This is the root cause of the mental health distress. Because you are distressed if you are in the middle of rising sea levels, or livelihoods being destroyed and all the rest of it. But you are equally as distressed if you are people like me watching it from afar, but actually, that is where our heart belongs.”
- “We are only asking from the vulnerable people or the people who have been distressed by this climate change and its impact. But we also need to ask the contributors, the larger contributors to the table, what they're doing to minimise the impact they're having on the climate.”
- “Most of the time, we see people go for the top-bottom approach, but we can have a bottom-up approach. We can sit down and have a bottom-up approach with the people who go through this mental suffering of climate change, and then we can discuss with them how to effectively do something so that their lives won't be hampered by climate change, and they will have a better way to settle things, as well as we could learn from that. We could also implement that through better finance and technical support to them.”
- “Climate change tends to be scientific, economic, and also political. We can develop new technology, employ new businesses, and even invest in a strategy. As I said before, we get governments to enact new laws, all of which are essential.”
- “I think the most important challenge, when we think of climate change, is the knowledge or awareness of climate change among the community. So, when it comes to climate change and health, health literacy in relation to climate change among populations, I think, is lacking. So, we

need to find means and mechanisms to ensure climate risk communication, so we can get people more engaged in climate-related activities.”

- “I think the climate discourse from the climate change players has been evolving for 25 years, somewhat separately from the public health space. And I think our challenge has been to bring these communities of donors and actors together, which is not easy at the national level. This is our learning and our challenge. The climate adaptation discourse and the global health community need to be in greater sync, going forward.”
- One of the pieces that is very important to understand is, what are the different factors, or domains of factors, that tend to impact mental health a lot more. So, for instance, apart from the social-economic challenges, there can also be loss of identity, loss of cultural identity, especially for tribal communities, we’ve seen, which might impact mental health a lot more and especially for migratory population, that’s going to be quite critical.”
- “Weather patterns are going to keep changing and evolving, and people will also keep learning to adapt to the mental stress and anxiety that comes with that. Because there is so much uncertainty around it, this needs to be captured longitudinally. It’s not a one-time effort and then you can go into programme design. That was another thing that is a challenge, that we should probably be overcoming.”

Glossary of terms

For a glossary describing relevant concepts and keywords for the Connecting Climate Minds research and action agendas, please download from [here](#).

Who produced this report

Author contribution statement

- Sameen Nasar¹: Project Administration, Investigation, Data Curation, Formal Analysis, Writing - Original Draft, Writing - Review and Editing
- Wafa Alam¹: Project Administration, Investigation, Data Curation, Formal Analysis, Writing - Original Draft, Writing - Review and Editing
- Md Tanzirul Alam¹: Investigation, Data Curation, Formal Analysis, Writing - Original Draft.
- Aishath Enash^{2, 3}: Writing - Review and Editing
- Poornima Prabhakaran⁴: Writing - Review and Editing
- Kunnuru Duishobekova⁵: Writing - Review and Editing
- Kezang Wangmo⁶: Writing - Review and Editing
- Inayat Singh Kabir⁷: Writing - Review and Editing
- Jawan Shir Rashik⁸: Writing - Review and Editing
- Dr Farzana Misha¹: Supervision, Investigation, Formal Analysis, Writing - Original Draft, Writing - Review and Editing

¹ BRAC James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

² U-Inspire Maldives

³ Ministry of Climate Change, Environment and Energy, Maldives

⁴ Centre for Health Analytics Research and Trends, Trivedi School of Biosciences, Ashoka University, India

⁵ Kyrgyz-Turkish University Manas

⁶ BRAC University

⁷ The George Institute for Global Health | INDIA

⁸ Afghanistan Research and Evaluation Unit (AREU)

This is the independent work of the authors with the support and input of the Connecting Climate Minds team.

Acknowledgements

The authors would like to thank dialogue participants, facilitators and notetakers for their contributions. Dialogues participants include:

- Aditya Dev Sood (Dialogue 2, Post Dialogue Survey)
- Amar Rathod (Dialogue 2, Post Dialogue Survey)
- Ahmed Hasib Bulbul (Dialogue 1, Post Dialogue Survey)
- Aishath Enash (Dialogue 1, Post Dialogue Survey)
- Ajwah Nadeem (Dialogue 1, Post Dialogue Survey)
- Ariful Islam (Dialogue 1, Post Dialogue Survey)
- Ananna Deb (Dialogue 2, Post Dialogue Survey)
- Anika Tasnim (Dialogue 2, Post Dialogue Survey)
- Aytach Meretliyeva (Dialogue 2, Post Dialogue Survey)
- Azizun Nahar (Dialogue 1, Dialogue 2 And Post Dialogue Survey)
- Bidhan Chandra Pal (Post Dialogue Survey)
- Bhawana Shrestha (Dialogue 1, Post Dialogue Survey)
- Dana Akhmetzhanova (Dialogue 1, Post Dialogue Survey)
- Daanish Mohamed Ali (Dialogue 2, Post Dialogue Survey)
- Dauod Khuram (Dialogue 1, Post Dialogue Survey)
- Dikshya Subedi (Dialogue 1, Post Dialogue Survey)
- Divya Bhardwaj (Dialogue 2, Post Dialogue Survey)
- Diyanath Samarasinghe (Dialogue 1, Post Dialogue Survey)
- Dr. A.M.A. Saja (Dialogue 2, Post Dialogue Survey)
- Dr. Md. Moudud Ahmed (Post Dialogue Survey)
- Elena Alexeyeva (Dialogue 1, Post Dialogue Survey)
- Emran Hasan (Dialogue 1, Post Dialogue Survey)
- Faria Kabir (Dialogue 2, Post Dialogue Survey)
- Farida Hafi (Dialogue 2, Post Dialogue Survey)
- Fariha Haidary (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Farzana Manzoor (Post Dialogue Survey)
- Hamidah Ashrafi Fateha (Dialogue 1, Post Dialogue Survey)
- Hasib Masum (Dialogue 2, Post Dialogue Survey)
- Hrushitaa Murali (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Ihsanullah Ghafoori (Dialogue1, Dialogue 2, Post Dialogue Survey)
- Inayat Singh Kakar (Dialogue1, Dialogue 2, Post Dialogue Survey)
- Jawan Shir Rasikh (Dialogue1, Post Dialogue Survey)
- Jannatul Mouwa (Post Dialogue Survey)
- K.M. Miraj Sujoy (Dialogue1, Post Dialogue Survey)
- Kezang Wangmo (Dialogue1, Dialogue 2, Post Dialogue Survey)
- Kunnuru Duishobekova (Dialogue1, Post Dialogue Survey)
- Khondker Zakiur Rahman (Dialogue 2, Post Dialogue Survey)
- Lahiru Charith Kodituwakku (Dialogue1, Post Dialogue Survey)
- Lailuma Kakar (Dialogue 2, Post Dialogue Survey)
- Lilanga Hasini Fernando Batawalage (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Manasi Gupta (Dialogue1, Post Dialogue Survey)
- Mariyam Shizna (Dialogue 1, Post Dialogue Survey)
- Shoeb Chowdhury (Dialogue 2, Post Dialogue Survey)
- Maruf Hossain Mishuk (Dialogue1, Post Dialogue Survey)
- Matrika Devkota (Dialogue 1, Dialogue 2)
- Maftuna Manapova (Post Dialogue Survey)
- Maya Rani Roy (Dialogue 1)

- Md Al Mamunur Rashid (Dialogue 2, Post Dialogue Survey)
- Md Nurul Islam (Dialogue 2, Post Dialogue Survey)
- Md Sohanur Rahman (Dialogue 2, Post Dialogue Survey)
- Md. Reza-A-Rabby (Post Dialogue Survey)
- Minal Karani (Dialogue 2, Post Dialogue Survey)
- Md. Humayun Kabir (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Md. Taifur Islam (Dialogue 1, Post Dialogue Survey)
- Mohaimin Ul Kabir (Dialogue 2, Post Dialogue Survey)
- Mokaram Hossain (Dialogue 2, Post Dialogue Survey)
- Mohammad Budrudzaman (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Mokhlesur Rahman (Dialogue 2, Post Dialogue Survey)
- Md. Nurul Islam Biplob (Dialogue 2, Post Dialogue Survey)
- Naurin Nur Sabah (Dialogue 2, Post Dialogue Survey)
- Nibras Nilayum Reza (Dialogue 2, Post Dialogue Survey)
- Nidhi Nagabhatla (Dialogue 2, Post Dialogue Survey)
- Rowsan Ara (Dialogue 2, Post Dialogue Survey)
- Monira Rahman (Dialogue 1, Post Dialogue Survey)
- Monuja Akter (Dialogue 1)
- Mostafa Hadei (Dialogue 1, Post Dialogue Survey)
- Muhammad Abdur Rahaman (Dialogue 1, Post Dialogue Survey)
- Muhammad Riaz Hossain (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Nadia Barki (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Nandita Bhatla (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Nazia Islam (Dialogue 1, Post Dialogue Survey)
- Nim Lhamo (Dialogue1, Post Dialogue Survey)
- Nimananda Rijal (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Payal Kumari Sah (Dialogue1, Post Dialogue Survey)
- Poorvaprabha Patil (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Rania Rajput (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Sabina Faiz Rashid (Dialogue 1)
- Sadril Shajahan (Post Dialogue Survey)
- Sadek Ahemmed (Dialogue 2, Post Dialogue Survey)
- Sadia Jahan Rothi (Dialogue 2, Post Dialogue Survey)
- Saifa Raz (Dialogue 2, Post Dialogue Survey)
- Sameera Zaman (Dialogue 1, Post Dialogue Survey)
- Saraf Mashiat (Dialogue 2, Post Dialogue Survey)
- Shahek Baloch (Dialogue 1, Post Dialogue Survey)
- Shamoel Haque (Dialogue 2, Post Dialogue Survey)
- Mohamed Alif Arif (Dialogue 1, Post Dialogue Survey)
- Sharmin Nahar Nipa (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Shubha Murthi (Dialogue1, Post Dialogue Survey)
- Shumaila Perveen (Dialogue 1, Post Dialogue Survey)
- Shumayta Marzia (Dialogue 2, Post Dialogue Survey)
- Soumyadip Dewanji (Post Dialogue Survey)
- Soyed Asadul Haque (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Sumaia Akter Sathi (Dialogue 2, Post Dialogue Survey)
- Tashfiya Zaman Mouree (Dialogue 2, Post Dialogue Survey)
- Tahura Farbin (Dialogue 1, Post Dialogue Survey)
- Syed Shabab Wahid (Dialogue 1, Post Dialogue Survey)
- Sz Opu (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Tapas Ranjan Chakraborty (Dialogue 1, Dialogue 2, Post Dialogue Survey)
- Uma Sunder (Dialogue 1, Post Dialogue Survey)
- Urvita Bhatia (Dialogue1, Post Dialogue Survey)
- Vanee Surendranathan (Dialogue 1, Post Dialogue Survey)
- Vishakha Joshi (Dialogue 1, Post Dialogue Survey)
- Wais Aria (Dialogue 1, Post Dialogue Survey)
- Wameq Raza (Dialogue 1, Dialogue 2, Post Dialogue Survey)

- Waqas Hameed (Dialogue 1, Post Dialogue Survey)
- Zul Merali (Dialogue 2, Post Dialogue Survey)
- Бегим Жанузакова (Dialogue 1, Post Dialogue Survey)

Facilitators:

Dialogue 1 and 2:

- Selima Sara Kabir
- Syeda Tahmina Ahmed
- Ishrat Jahan
- Sabiha Mahbub Khan
- Tahsin Madani
- Ranjan Koiri

Notetakers:

Dialogue 1 and 2:

- Antara Roy
- Israr Hasan
- Nafisa Benazir Bhuiyan
- Riya Siddiqua
- Maliha Mehnaz Mitu
- Amal Chowdhury
- Ishrat Binte Aftab
- Yasmin Ara
- Swaksar Adhikary
- Tahia Hasan
- Mahmudul Hasan Shesheir
- Ribica Chowdhury
- Aushmita Akbar
- Tahia Hasan
- Mahmudul Hasan Shesheir

Funding

The project was funded by Wellcome.

Conflicts of interest

The authors have no conflict of interest to declare.

Appendix

Appendix 1: Dialogue agendas

Dialogue 1

Time (mins)	What	How
0-20	Welcome and introductions	
20-35	Visioning Exercise	Visioning Exercise: “Headlines” on Climate Centre Good Games platform Participants share what they are hoping to get from being part of Connecting Climate Minds, and why they are invested in this topic <i>Imagine we are in 2030 and the CCM project has received an important prize for its achievements in bringing CC/MH together in your region and making change for climate and mental health together through research and action. You are interviewed by a journalist as you are celebrating winning the prize.</i> Create / vote on and review Headlines created on the Good Games Platform
35-50	Scene setting	Brief context to the links between climate change and mental health What are we hearing in the region that we can come together and respond to: sharing highlights of pre-scoping Presentation on Connecting Climate Minds and Q&A
50-95	Breakout session 1: Regional needs	Host sets scene for breakout discussion Share climate hazards overview for the region Breakout groups Facilitator to lead the group in a discussion on the mental health challenges arising from climate change in the region. Questions posed to participants were: • <i>From your perspective, what are the climate hazards that concern you the most in terms of mental health and wellbeing?</i> • <i>For each hazard, discuss the following:</i> ○ <i>Who is being impacted? Are there any groups more at risk?</i> ○ <i>What are the mental health and wellbeing outcomes associated with this hazard?</i>

		○ How does this hazard specifically affect mental health and wellbeing; what are the mechanism e.g. housing, livelihoods etc ○ Encourage people to look at different types of mechanisms e.g. social, biological, cultural, environmental ○ Climate adaptation/mitigation actions - what is being done, what can be done? ○ Actions that are being/could be taken to improve or promote mental health and wellbeing in this context Brief plenary highlights
95-105	Break	
105-110	Sneak preview	The cartoon artist shares one first draft reflecting some of the content of the session so far. Host invites participants to share in the chat how the image resonates.
110-160	Breakout session 2: Evidence gaps and research priorities	Breakout groups [same participants, facilitator and note-taker as first breakout session] Looking back at the hazard table from Breakout session 1 as a starting point, participants are guided to start thinking about where more research is needed. Hosts invite 2 or 3 facilitators to share some brief reflections
160-170	Cartoon reflection on dialogue output	Participants individually review each cartoon and share their reflections. This can range from: suggestions for changes to make it more accurate, to sharing how it relates to our reality, or any other insights.
170-180	Wrap-up and next steps	Host shares a high-level recap of the session, thanks everyone and celebrates this first coming together of the regional community. Participants complete brief survey to provide feedback on the dialogue

Dialogue 2

Time (mins)	What	How
0-5	World Map Exercise	Host asks participants to place a “star” icon where they currently are and then take the “heart” icon and place it where they would like to visit one day or a place they have been and really loved/feel especially connected to.

5-15	Welcome and introductions	
15-30	Sharing updates	Presentation on global and regional progress and outcomes of dialogue 1, including vision, priority research themes.
30-50	Priority Research Theme Feedback	Jamboard shared on screen. Host introduces priority research theme activity and asks people to consider the following questions: Research questions • <i>Is this a valuable theme? Would you add or change anything?</i> • <i>Share any ideas on datasets, metrics or methods that could be applied to help address this.</i> Research topics • <i>What would be most helpful to know about this topic to understand and respond to it?</i> • <i>What would you want to know first?</i> Music plays and participants work independently on Jamboard at their own speed.
50-55	Scene setting for breakouts	Host sets the scene for the breakout discussion. Interested in the 'how', which includes the principles and desired outcomes for research, policy, and practice in the region. The discussion will enable targeting of action and investment to both implement the research agenda in the region, and to ensure evidence on the climate and mental health nexus is translated into policy and practice
55-90	Break out 1	Eight breakout groups. The first 4 on the theme <i>“Creating knowledge through research”</i> (e.g., How can we best implement the research agenda?) and the other 4 on <i>“Fostering evidence-based policy and action”</i> (e.g. How can we translate the growing evidence base to action in policy and practice?). Each group uses a Jamboard to discuss: 1) What does this look like when it is done well? 2) What opportunities & enablers exist? 3) What challenges must we overcome? 4) Who should our partners and stakeholders be? Brief plenary highlights
90-100	Break	
100-105	Draw your feelings	Host welcomes everyone back and invites all who'd like to join to grab a piece of paper and a pen/marker (or digital) - and draw how they are feeling today. Anyone willing is invited to show their image - share some reflections.

105-125	Spectrum Mapping	Host introduces and explains the activity. Share screen of Jamboard. Designed to reveal the diversity of perspectives and options around different concepts and to organise them into a meaningful spectrum. This activity aims to illuminate the group's range of perspectives and to organise those perspectives into a continuum. Question: <i>What do these concepts mean to you?</i> Focusing on one board at a time, host asks everyone to silently generate a point-of-view around that concept and write it on a sticky note on the Jamboard. People can add more than one. Once the sticky notes are posted, the host works with the group to sort them into a horizontal range of ideas. Sticky notes that express similar perspectives or options should go next to each other. Sticky notes that seem to be outliers should stand alone; they may sometimes end up defining the limits of the range. Repeat with the remaining concepts. At the end, ask for observations and insights on the lay of the land. And if any perspective or option has been excluded. If so, add it and re-sort as necessary.
125 - 155	Break out 2	Host introduces the second round of breakout rooms and provides a 2-minute summary of the Jam board contributions from the first breakout. Original 8 groups reconvene and expand and add to their initial thoughts. Groups are allowed to continue on the theme they were already discussing or switch to the other theme.
155-175	I will ... I want...	Host asks everyone to personally reflect about how they will contribute to this space and share an "I will" statement in the chat, but only press enter when told to do so. After a minute, host invites everyone to press enter at the same time to make statements appear. This is repeated with a statement about "Out of this community I would like to see..."

175	Wrap up and next steps	Host shares a high-level recap of the session, thanks everyone and outlines next steps. Participants complete brief survey to provide feedback on the dialogue
-----	-------------------------------	--

Appendix 2: Dialogue methodology

1. Coding framework for research agenda

Research category	Sub-categories
1. Impacts, risks and vulnerable groups This category is about improving our understanding of the ways in which mental health is affected by climate change. For example, what mental health outcomes are impacted or at risk, the prevalence, severity, economic and societal cost of these impacts, and who is more vulnerable to these impacts. This category also includes the ways we can go about getting this improved understanding of mental health impacts of climate change - the methods and metrics we need to assess and monitor mental health in ways that are relevant to climate change, contextually appropriate, comparable etc.	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> Timeframe Geographical variation
	1.1. Research that focuses on the prevalence, severity and nature of the experience of different mental health outcomes/challenges/experiences affected by different aspects of climate change. This may include research to understand the emergence of climate-specific mental health experiences and their relationship to already defined mental health challenges.
	1.2. Quantifying the fraction of mental health burden (including mortality) caused by climate change.
	1.3. Understanding the risk factors to mental health that are caused or affected by climate change as well as protective factors.
	1.4. Identifying population sub-groups (e.g. demographics, livelihoods, life stage, pre-existing mental health challenges) who experience increased vulnerability to mental health challenges caused by climate change, and conversely those experiencing resilience to these effects (i.e. vulnerable groups).
	1.5. Quantifying the cost (e.g. economic, social) of the additional mental health burden caused by climate change and insufficient climate action.
	1.6. Methods research to identify the most appropriate ways to assess and monitor the mental health impacts of climate change [including adapting pre-existing scales, creating new ones, determining appropriate mental health metrics and indicators for inclusion in global processes like Lancet Countdown]. This can also include the need for cross-cultural validation and development of culturally appropriate metrics.

2. Pathways and mechanisms This category is about improving our understanding of <i>how</i> mental health is affected by climate change. We are interested in research themes that can help identify, categorise and understand the range of ways that climate change or climate action may act to affect mental health. This can include considering pathways and mechanisms that are biological, psychological, societal or environmental in nature, and may be direct or indirect. Note that mechanisms can include mechanisms to the development, maintenance, and/or resolution of mental challenges, so this includes also mechanisms relevant to guide development or understand workings of interventions	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> What factors are linked with increased vulnerability or increased resilience for the associated mental health outcomes. 2.1. Categorising and understanding the societal mechanisms by which climate change negatively impacts mental health [e.g. changes to livelihoods, disruption to cultural practices, food and water insecurity, forced migration, political factors] 2.2. Categorising and understanding the environmental mechanisms by which climate change negatively impacts mental health [e.g. air pollution, reduced exposure to biodiversity] 2.3. Categorising and understanding the psychological mechanisms by which climate change negatively impacts mental health [e.g. how temperature affects cognitive changes relevant to symptoms of mental health challenges]. 2.4. Categorising and understanding the biological mechanisms by which climate change negatively impacts mental health [e.g. impacts of psychotropic medication on thermoregulation, neurodevelopmental factors]. 2.5. Understanding mechanisms whereby climate action or mental health interventions benefit climate and mental health (i.e. co-beneficial mechanism). 2.6. Methods research to identify the most appropriate ways to assess and monitor pathways and mechanisms by which climate change negatively impacts mental health and wellbeing (e.g. systems mapping across disciplines)
3. Mental health benefits of climate action (adaptation and mitigation) This category is about how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental	3.1. Identifying climate actions that integrate or align with mental health benefits [co-beneficial climate actions, e.g. increased tree cover in cities] 3.2. Quantifying co-benefits of climate action for mental health (including number of people experiencing the benefit, size of effect, economic considerations). 3.3. Exploring how the mental health costs and benefits of climate action may differ across population sub-groups (e.g. demographics, livelihoods, life stage) 3.4. Understanding the governance structures/decision support tools that enables alignment of action for climate change and for mental health across sectors 3.5. Mapping and monitoring the integration of mental health within adaptation and mitigation policies across sectors [e.g., National Adaptation Plans, energy, transport,

health. This includes quantifying costs and benefits of climate action for mental health, understanding what is needed to support better alignment between climate action and mental health action, and identifying where this integration is already happening across strategies and policies.	food, water, agriculture]
	3.6. Exploring opportunities for mental health to be integrated into other climate priority areas i.e., loss and damage and climate finance.
	3.7. Determining best approaches for climate action (e.g., emissions reductions or climate adaptation) within the mental health sector (ensuring psychiatric facilities can be kept cool in heat waves; green space projects in mental healthcare facilities)
	3.8. Methods research to identify the most appropriate ways to assess and monitor mental health benefits of climate action [e.g., place-based approaches, methods for attributing and quantifying co-benefits, methods for assessment of the mental health implications of decisions in other sectors]
4. Mental health interventions/solutions in the context of climate change This category is about identifying the most effective mental health interventions/solutions to support mental health in the context of climate change. This might be about providing support to people already experiencing negative mental health impacts, or about reducing risk or severity of future negative mental health impacts. This includes learning from knowledge held by different disciplines, communities and cultures, understanding how existing mental health interventions are affected by climate change, identifying and evaluating existing interventions that are relevant to the context of climate	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> LEVEL (e.g.) Individual, Family, Community, Systems MECHANISM (e.g.) Biological, Psychological, Social, Environmental SECTOR (e.g.) Education, Healthcare, Public Health Effectiveness considerations include impacts across different population groups, and implementation considerations might include providers, cost and time.
	4.1. Understanding different ways of knowing, being and doing in different cultures and communities that can build individual, community and ecological resilience
	4.2. Understanding how existing mental health interventions are affected by climate change
	4.3. Identifying and evaluating mental health interventions that are already designed for or relevant to the context of climate change and/or integrate climate change considerations
	4.4. Amending, implementing and evaluating relevant mental health interventions from other settings to be appropriate for climate-related impacts?
	4.5. Co-designing, implementing and evaluating novel interventions that meet climate-related mental health needs
	4.6. Comparing cost-effectiveness, implementation considerations, and effectiveness across interventions for a particular setting and particular population group to determine "best buys"
	4.7. Identifying, developing and evaluating approaches to awareness-raising and capacity building to upskill workforces to recognise and respond to the mental health

change, and developing new interventions. Interventions are relevant at all levels (individual, family, community, systems) and across sectors.	impacts of the climate crisis (e.g. mental health professionals, emergency responders)
---	--

2. Coding framework for action agenda

Action category	Sub-category
1. Creating an enabling environment for research at the intersection of climate change and mental health	1.1 Desired state of research - This code captures <i>what good looks like</i> for climate change and mental health research in the region that implements the research agenda. - What are the features of the kind of research that is desired or valuable? Are there specific attributes or milestones that would signify this state of research?
	1.2 Opportunities and enablers This code captures opportunities to progress the climate and mental health research field in the region towards the desired state, and factors that would enable progress. May be general or specific and may link to what is required to overcome the challenges outlined in the next code.
	1.3 Challenges holding back research This code captures challenges that are stopping the climate and mental health field in the region from currently being in the desired state or are predicted to emerge in trying to create investment in and implementation of the research agenda.
	1.4 Partners/stakeholders This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for implementation of the research agenda in the region and securing required investment.
	1.5 Priority next steps/recommendations to investors and actors - This code captures the concrete next steps that need to be taken as priorities to create the conditions in the region to implement the research agenda. - This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.
2. Translating a growing evidence base into action that can respond to the mental health impacts of climate change	2.1 Desired state of evidence to action in policy and practice - This code captures <i>what good looks like</i> for action on climate change and mental health in the region based on current and future evidence. - What are the features of the kind of pathways for translating evidence into action that are desired or valuable? Are there specific attributes or milestones that would signify that evidence-based action is occurring?
	2.2 Opportunities and enablers This code captures opportunities to progress evidence-based action on climate and mental health in the region towards the desired state, and factors

	that would enable progress. May be general or specific and may link to what is required to overcome the challenges outlined in the next code.
	2.3 Challenges holding back action This code captures challenges that are stopping desired actions to protect mental health from the climate crisis in the region, or to enable co-beneficial climate action. The code may also include challenges that are predicted to emerge in trying to ensure that current and future evidence translates into change on the ground and at all levels of policy and practice.
	2.4 Partners/stakeholders This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for translation of evidence into relevant action and securing required investment.
	2.5 Priority next steps/recommendations to investors and actors - This code captures the concrete next steps that need to be taken as priorities to translate the emerging evidence base on climate and mental health into action in policy and practice. - This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.

Appendix 3: Survey methodology

The pre-dialogue survey was intended to gather demographic information about the participants, their awareness of, and personal or professional experience with climate change and/or mental health issues. Accessibility requirements were also asked to ensure that dialogues could be designed to suit the needs of participants. The post-dialogue survey was intended to (a) collect additional feedback on the draft research topics after they had been reviewed on the basis of discussions held during Dialogue 2, and (b) collect feedback on the project as a whole. Both surveys were implemented on Qualtrics, and the link was distributed via email.

The pre-dialogue survey was sent to dialogue participants only, while the post-dialogue survey (sent out after Dialogue 2) was sent to both dialogue attendees and the wider regional community, composed of individuals who had expressed interest in receiving project updates. Survey recipients were also asked to forward the post-dialogue survey to others within relevant networks. For those who had not attended the dialogue(s), the invitation to complete the survey included a Participant Information Sheet and a digital consent form. The dialogue participants had already consented as part of the dialogue process.

Appendix 4: Survey participant communications

An email was sent to all invited participants with the link to the survey and consent form (both to be filled out via Qualtrics). A participant information sheet was also shared in the email with information on what is required from the participants for the surveys and dialogues. Participants were only provided with the Zoom link to dialogue 1 upon completion of the survey, and the consent form. There were a few lived-experience participants who could not fill out the survey on their own due to language barriers (since the survey was only administered in English). We scheduled 20 - 30 minutes long online meetings with them on Zoom and helped fill out the survey for them.

The survey participants were requested to share the survey with people in their networks who work in this area in the region. The purpose of the post-dialogue survey was to seek feedback on the Priority Research Themes identified from the two dialogues and get their perception of the overall Connecting Climate Minds Project. Participants who were not attendees of either dialogue were also sent the participant information sheet and consent forms.

Appendix 5: Data collection and storage

Dialogues

Dialogues were conducted virtually on Zoom. Dialogues and breakout groups were recorded and transcribed by third-party providers (Way with Words and Absolute Translations) and Zoom chat comments were saved. In dialogue 1, Word documents were used to capture notes from breakout discussions. In dialogue 2, Google Jamboard was used to capture notes and for participants to directly contribute comments.

Surveys

Survey distribution and data collection were carried out using the online platform Qualtrics. All survey data was collected by Imperial College London and anonymous data was shared with BRAC JPGSPH for analysis.

Data storage and sharing

Data was stored and managed by Imperial College London using a secure server. BRAC JPGSPH was a Joint Data Controller for the data provided to this project for CSA and was responsible for securely storing and sharing data with Imperial College London and with regional analyst teams. Data will be stored by Imperial College London for 10 years after study completion.

Appendix 6: Climate hazards in Central and Southern Asia^{21, 22, 35}

Observed climate-related hazards in CSA and projected changes.

To guide informed discussions on the current and potential mental health consequences of current and future climate change in the region, it was vital to ensure a grounding in climate science. The Red Cross Red Crescent Climate Centre (RCCC) conducted a mapping of previous and projected climate hazards across the region to inform the dialogues and research agenda. The following is a summary of the climate-related hazards that Central and Southern Asia have experienced over the last 30 years and their projected increases to approximately 2030.

It is important to emphasise that the data gathered for the past and future periods differ not only in spatial resolution but are also based on different underlying principles. The former is through observations or documented occurrences such as those reported by national weather services and are aggregated at a country scale. The latter are from climate modelling-based studies and assessment reports published by the Intergovernmental Panel on Climate Change (IPCC), which are at a more granular spatial scale. For the same reason, the categories of climate-related hazards in the past and future periods may sometimes not be identical but are generally comparable or related. For instance, flooding that is reported as a relevant climate hazard in the last 30 years here is not a direct output of

global climate models (GCMs). Instead, the underlying meteorological variable (rainfall) from the simulations of GCMs are used for deriving indices that can be considered as a proxy for potential flooding in a future warming climate.

Central and Southern Asia

As a region, CSA is particularly vulnerable to climate change. This region has frequently experienced droughts, wildfires, floods, landslides, storms and extreme temperatures over the recent years, as international disaster database EM-DAT (<https://www.emdat.be/>) sources show. Note that the limited number of extreme events recorded (e.g. droughts, wildfires and extreme temperatures) may be due to the missing records in the EM-DAT data for various reasons, such as poor or limited network of satellite and surface observations over the region, or limited reporting from national meteorological services. In addition, the spatial aggregation of the data to country-scales makes it difficult to map the occurrence of the past hazards to human settlements. The total number of people affected as reported in EM-DAT therefore may not always be truly representative of the frequency of occurrences of the underlying climate hazard. For instance, a country in CSA may have experienced frequent heatwaves, but if these events were not recorded or in uninhabited regions, the total number of occurrences discussed below may be an underreporting.

Previous Droughts

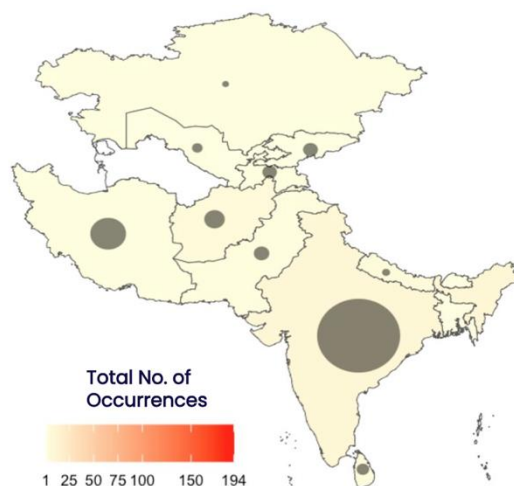
Although droughts and wildfires did not occur the most frequently of all climate-related hazards in Central and Southern Asia, overall, these events affected the most people with 138 million people in India, 21 million people in Iran and 5 million people in Afghanistan for example (Figure 3 top).

Drought and Wildfire

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 500K
- 1M
- 5M
- >10M

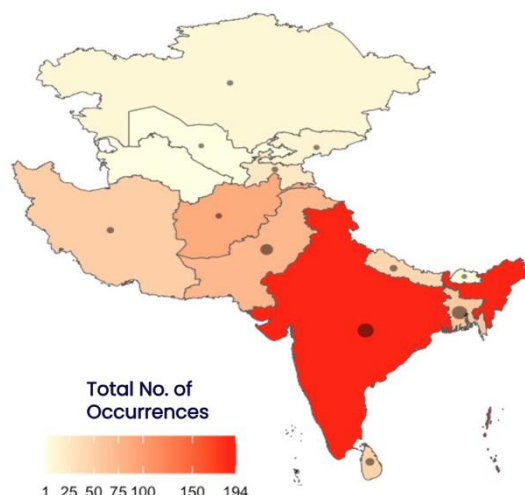


Flood and Landslide

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 500K
- 1M
- 5M
- >10M



Storm and Extreme Temperature

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 500K
- 1M
- 5M
- >10M

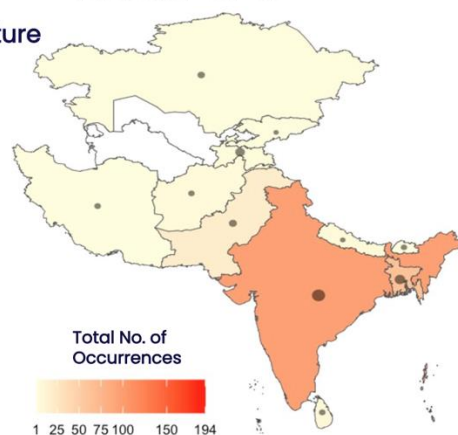


Figure 3: Number of reported droughts and wildfires (top), flood and landslides (middle) and storms and extreme temperatures (bottom) between 1991-2022. Red legend depicts the total number of occurrences of each category of hazard between 1991-2022, the black circle size is representative of the average affected population in that period.

Previous Floods

Occurrence of floods and landslides frequently take place in Central and Southern Asia, with 194 of such hydrological events arising in India, 86 events in Afghanistan and 73 events in Pakistan between 1991-2022 (Figure 3, middle). These events have affected up to 2.5 million people in India, 2.2 million people in Bangladesh and 1.1 million people in Pakistan.

Previous Extreme (Hot) Temperatures and Storms

Furthermore, India, Bangladesh and Pakistan are most affected by storms and extreme temperatures (Figure 3, bottom). 95 of such events occurred in India, 63 occurred in Bangladesh and 19 occurred in Pakistan, affecting up to 3 million people across these countries retrospectively.

Future Projections

Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the [Coupled Model Intercomparison Project - Phase VI \(CMIP6\)](#) multi-model

GCMs ensemble provided in the IPCC, 2021. While not the worst-case emissions scenario, SSP2-4.5 assumes that the 2015 Paris agreement commitments are not achieved and 2°C global warming is not avoided. GCMs have varying climate sensitivity and generally also differ in their representation of earth system processes. While their projections in the intensity of a particular hazard in the future may therefore differ despite a persistent signal in historical observed records, the direction of the signal (i.e., a positive or negative change of a climate hazard) are generally more robust across the GCMs.

- Mean and extreme temperatures: Future changes to the mean number of hot days above 35°C in 2030 compared to 1986-2005 are projected to increase particularly in the western parts of Central and Southern Asia (Figure 4a) (high confidence). These projections show an increase of more than 20 hot days in Iran, Afghanistan, Turkmenistan and southern India in 2030 (high confidence).
- Heavy precipitation events and flooding: The average change of annual maximum 5-day precipitation in the year 2030 relative to 1986-2005 is projected to increase heavy rainfall and pluvial flooding east of the Caspian Sea (high confidence), in Kazakhstan (high confidence) and India (Figure 4b) (high confidence).
- Sea level rise: Relative Sea level around Asia has increased faster than global average, with coastal area loss and shoreline retreat. Regional mean sea level will continue to rise (high confidence). This can be associated with more coastal flooding (high confidence).

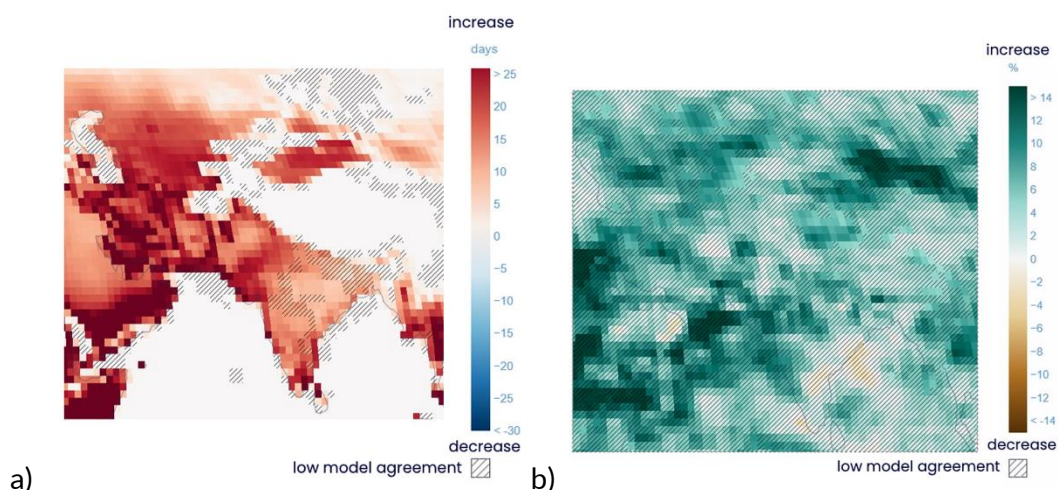


Figure 4: Projected changes in a) hot days above 35°C b) annual maximum 5-day precipitation for 2030 compared to 1986-2005 under the middle of the road emissions scenario SSP2-4.5.

Note: Detailed information on the projections and the associated confidence were accessed from the IPCC Working Group I contribution to the Sixth Assessment Report "Climate Change 2021: The Physical Science Basis" and the associated [IPCC WGI Interactive Atlas](#) and regional factsheets: [Fact Sheets | Climate Change 2021: The Physical Science Basis \(ipcc.ch\)](#)

References

- ¹ World Health Organization (WHO). *Mental health and climate change: policy brief*. (World Health Organization, 2022).
- ² Lawrance, E. L., Thompson, R., Newberry Le Vay, J., Page, L. & Jennings, N. The Impact of Climate Change on Mental Health and Emotional Wellbeing: A Narrative Review of Current Evidence, and its Implications. *International Review of Psychiatry* **34**, 443–498 (2022).
- ³ Roland, J., Kurek, N. & Nabarro, D. *Health in the climate crisis: A guide for health leaders*. (World Innovation Summit for Health, 2020).
- ⁴ Hickman, C. *et al.* Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey. *The Lancet Planetary Health* **5**, e863–e873 (2021).
- ⁵ Chen, S. X., Lee, M., McVea, D. A. & Henderson, S. B. Risk of mortality among people with schizophrenia during the 2021 heat dome. *British Columbia Medical Journal* **65**, 158–9.
- ⁶ Page, L. A., Hajat, S., Kovats, R. S. & Howard, L. M. Temperature-related deaths in people with psychosis, dementia and substance misuse. *Br J Psychiatry* **200**, 485–490 (2012).
- ⁷ Woodland, L., Ratwatte, P., Phalkey, R. & Gillingham, E. L. Investigating the Health Impacts of Climate Change among People with Pre-Existing Mental Health Problems: A Scoping Review. *IJERPH* **20**, 5563 (2023).
- ⁸ Charlson, F. *et al.* Climate Change and Mental Health: A Scoping Review. *IJERPH* **18**, 4486 (2021).
- ⁹ World Health Organization (WHO). *2021 WHO Health and Climate Change Survey Report*. (World Health Organization, 2021).
- ¹⁰ Romanello, M. *et al.* The 2022 report of the Lancet Countdown on health and climate change: health at the mercy of fossil fuels. *The Lancet* **400**, 1619–1654 (2022).
- ¹¹ Gale, N. K., Heath, G., Cameron, E., Rashid, S. & Redwood, S. Using the framework method for the analysis of qualitative data in multi-disciplinary health research. *BMC Med Res Methodol* **13**, (2013).
- ¹² Javed, A. *et al.* Reducing the stigma of mental health disorders with a focus on low- and middle-income countries. *Asian J Psychiatr* **58**, 102601 (2021).
- ¹³ Naveed, S. *et al.* Corrigendum: Prevalence of Common Mental Disorders in South Asia: A Systematic Review and Meta-Regression Analysis. *Front Psychiatry* **11**, (2020).
- ¹⁴ Hook, K. & Bogdanov, S. Mental health care in Eastern Europe and Central Asia: An analysis of needs and a call for greater investment. *Lancet Reg Health - Eur* **10**, 100182 (2021).
- ¹⁵ Aliev, A.-A. *et al.* Widespread collapse, glimpses of revival: a scoping review of mental health policy and service development in Central Asia. *Soc Psychiatry Psychiatr Epidemiol* **56**, 1329–1340 (2021).
- ¹⁶ Patra, S. Climate Change and Health Vulnerability in South Asia. *IJHSSI* **2**, 31–9 (2013).

- ¹⁷ Manning, C. & Clayton, S. Threats to mental health and wellbeing associated with climate change. in *Psychology and Climate Change* (eds. Clayton, S. & Manning, C.) 217–244 (Academic Press, 2018).
- ¹⁸ Anderson, C. A. Heat and violence. *Curr Dir Psychol Sci* **10**, 33–38 (2001).
- ¹⁹ Bharadwaj, R., Karthikeyan, N. & Deulgaonkar, I. *Urgent preventative action for climate-related suicides in rural India*. (IIED, 2023).
- ²⁰ Tasdik Hasan, M. *et al.* Exploring mental health needs and services among affected populations in a cyclone affected area in coastal Bangladesh: a qualitative case study. *Int J Ment Health Syst* **14**, (2020).
- ²¹ Intergovernmental Panel on Climate Change (IPCC). *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. (Cambridge University Press, 2023).
- ²² Knutson, T. *et al.* Tropical Cyclones and Climate Change Assessment: Part II: Projected Response to Anthropogenic Warming. *Bulletin of the American Meteorological Society* **101**, E303–E322 (2020).
- ²³ Aghababaeian, H. *et al.* Global Health Impacts of Dust Storms: A Systematic Review. *Environ Health Insights* **15**, (2021).
- ²⁴ WHO. What Are The WHO Air Quality Guidelines? WHO <https://www.who.int/news-room/feature-stories/detail/what-are-the-who-air-quality-guidelines> (2021).
- ²⁵ National Hydrometeorological Service of the Republic of Kazakhstan. Forecast for CO gas. *National Hydrometeorological Service of the Republic of Kazakhstan* <https://www.kazhydromet.kz/vc/silam/> (2024).
- ²⁶ Naveed, S. *et al.* Corrigendum: Prevalence of Common Mental Disorders in South Asia: A Systematic Review and Meta-Regression Analysis. *Front Psychiatry* **11**, (2020).
- ²⁷ Begum, A. *et al.* Prevalence of suicide ideation among adolescents and young adults in rural Bangladesh. *Int J Ment Health* **46**, 177–187 (2017).
- ²⁸ Lindseth, A. & Norberg, A. A phenomenological hermeneutical method for researching lived experience. *Scand J Caring Sci* **18**, 145–153 (2004).
- ²⁹ Frechette, J., Bitzas, V., Aubry, M., Kilpatrick, K. & Lavoie-Tremblay, M. Capturing Lived Experience: Methodological Considerations for Interpretive Phenomenological Inquiry. *Int J Qual Methods* **19**, 160940692090725 (2020).
- ³⁰ Majumder, J., Saha, I., Saha, A. & Chakrabarti, A. Climate Change, Disasters, and Mental Health of Adolescents in India. *Indian J Psychol Med* **45**, 289–291 (2023).
- ³¹ US EPA. Particulate Matter (PM) Basics. US EPA <https://www.epa.gov/pm-pollution/particulate-matter-pm-basics> (2023).
- ³² Roberts, S. *et al.* Exploration of NO₂ and PM_{2.5} air pollution and mental health problems using high-resolution data in London-based children from a UK longitudinal cohort study. *Psychiatry Res* **272**, 8–17 (2019).

- ³³ Xue, T. *et al.* Long-term PM2.5 exposure and depressive symptoms in China: A quasi-experimental study. *Lancet Reg Health West Pac* **6**, 100079 (2021).
- ³⁴ Hautekiet, P. *et al.* Air pollution in association with mental and self-rated health and the mediating effect of physical activity. *Environ Health* **21**, (2022).
- ³⁵ CRED. EM-DAT: The International Disaster database. <https://www.emdat.be/> (2023).